

RENEWABLE ENERGY SOURCES AND APPLICATIONS

(Subject Code: 21MEO112T)

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Unit-3 - Ocean, Hydro and Geothermal Energy

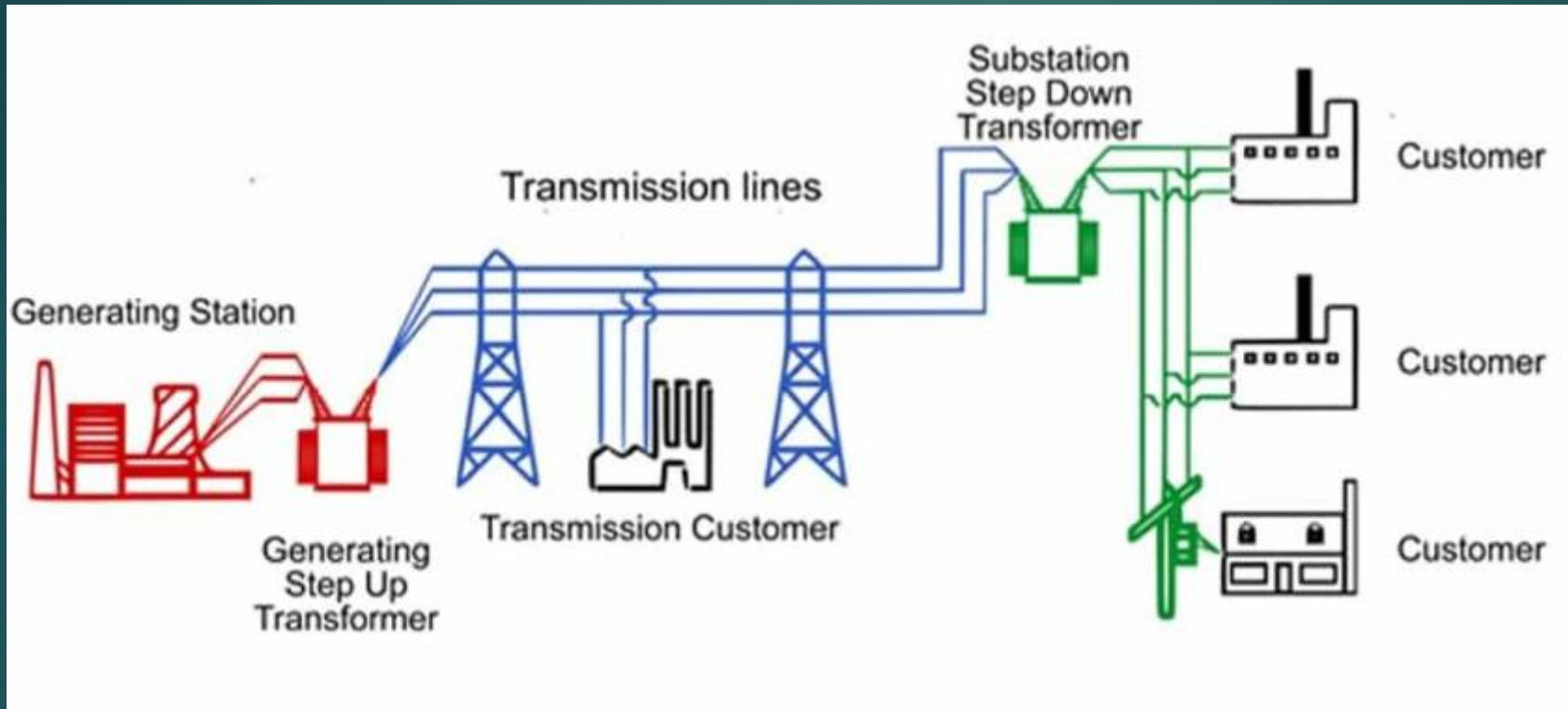
Unit-3 - Ocean, Hydro and Geothermal Energy

Current scenario of Ocean Thermal Energy Conversion (OTEC) system - open cycle & closed cycle, environmental impacts, challenges, tidal energy - single basin and double basin plants. Wave energy conversion systems, Geothermal energy: Origin, applications, types of geothermal resources. Hydroelectric power plant.

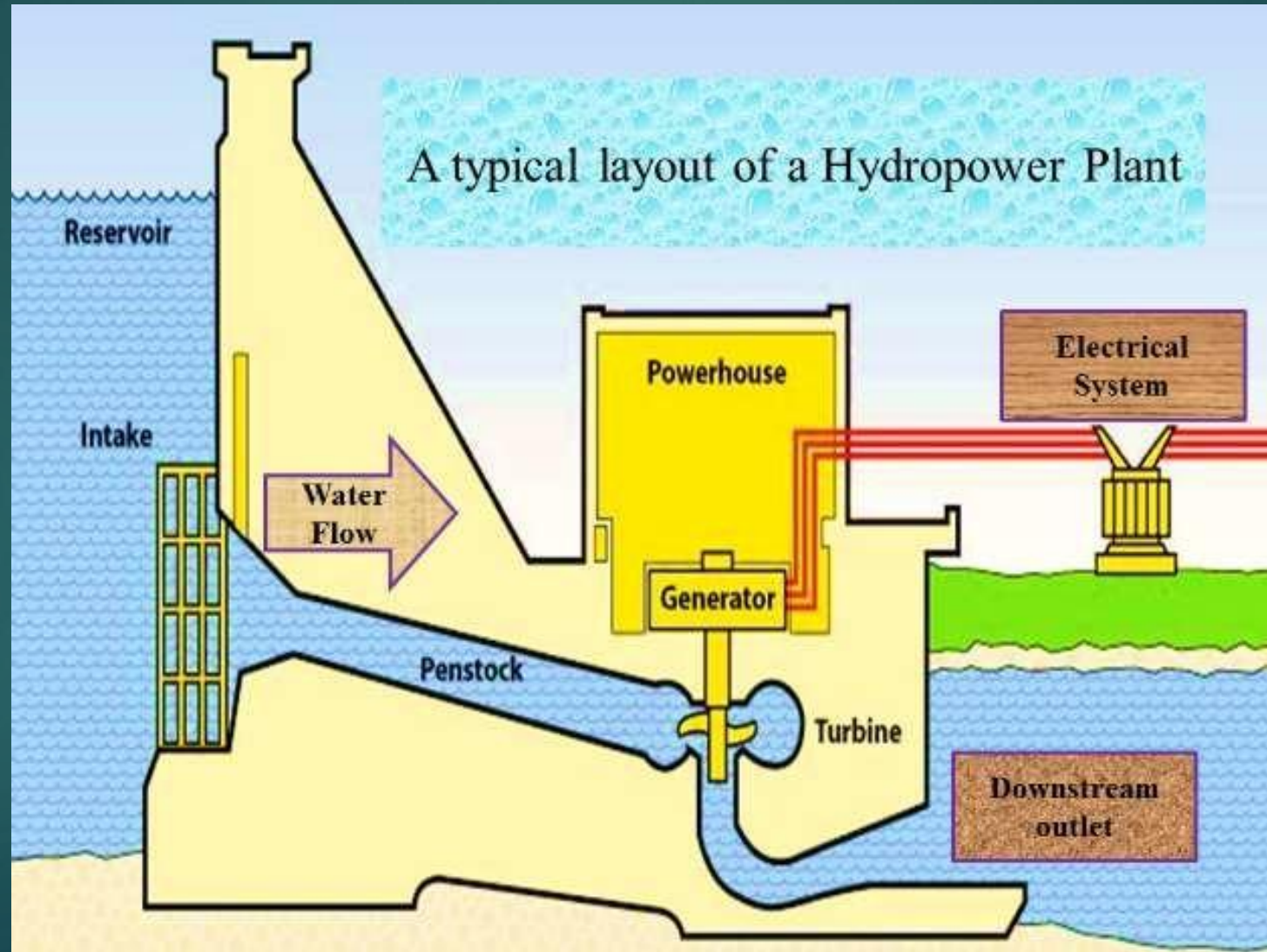
Dr. Amit Kumar

Hydroelectric power plant

How power is generated and distributed in hydropower plants?



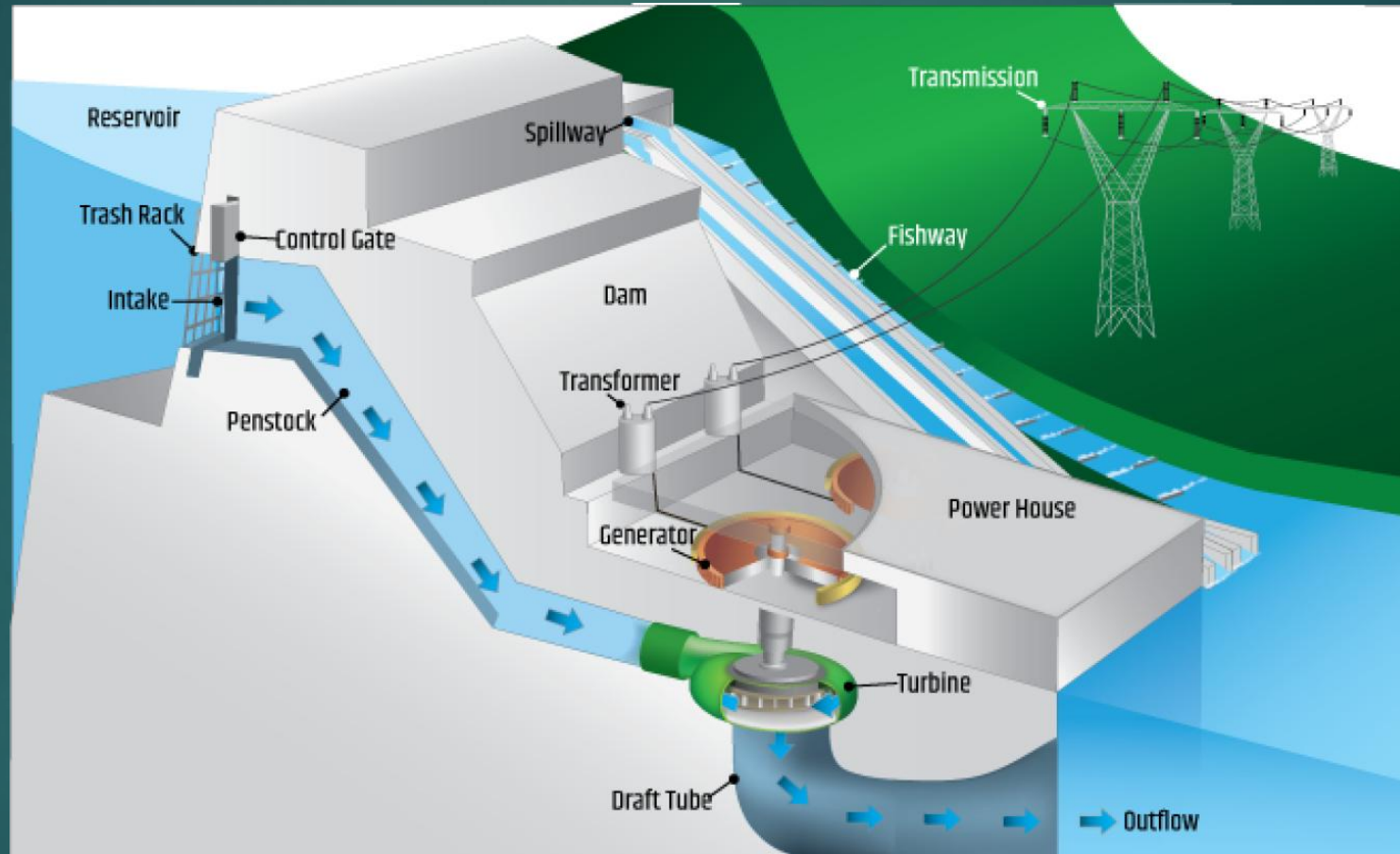
General Layout of Hydro-electric power plant



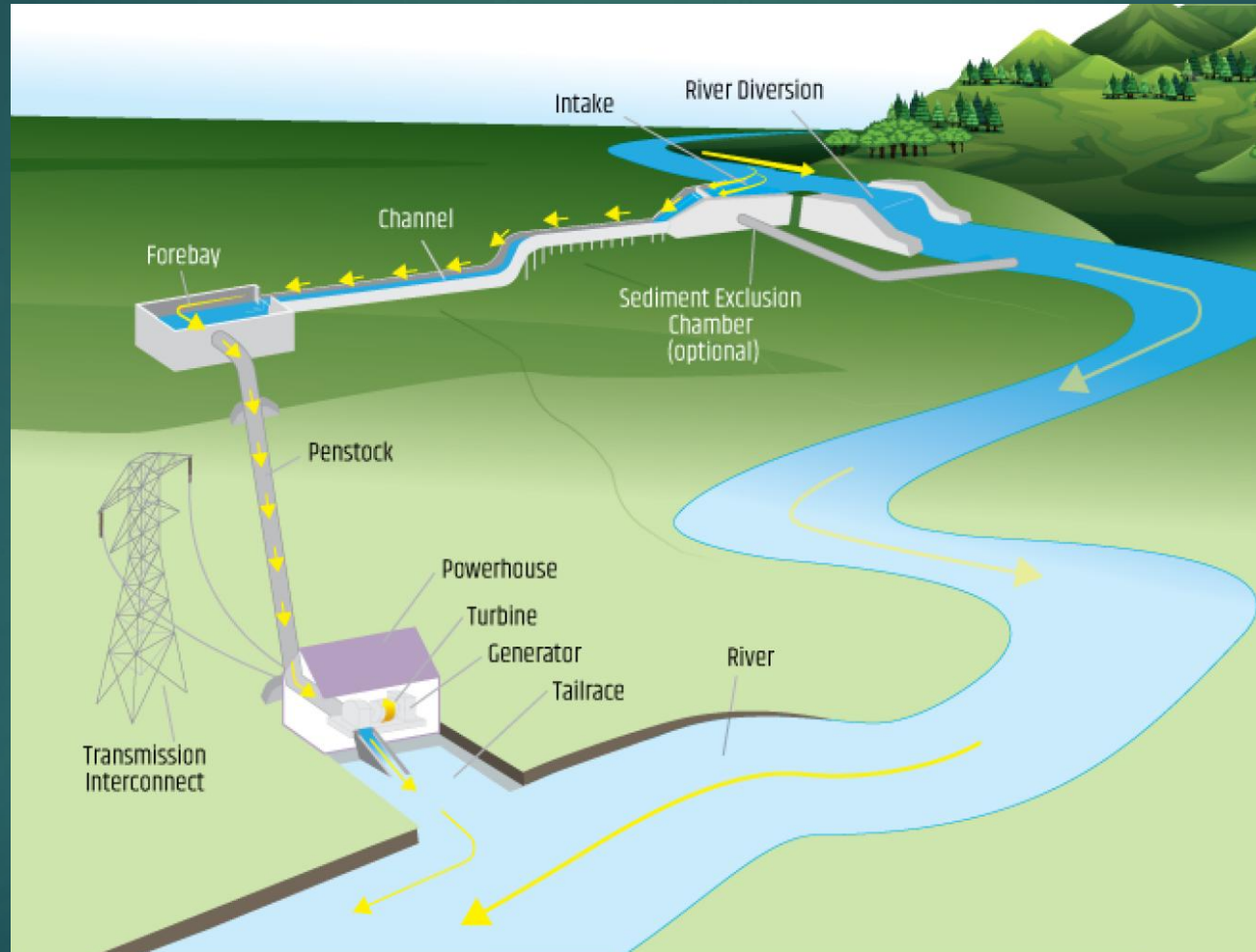
Types of Hydroelectric power plant

- Impoundment (uses a dam and reservoir)
- Diversion (uses a weir and channel to divert river flow)
- Pumped Storage (uses two reservoirs to store and release water for energy storage)

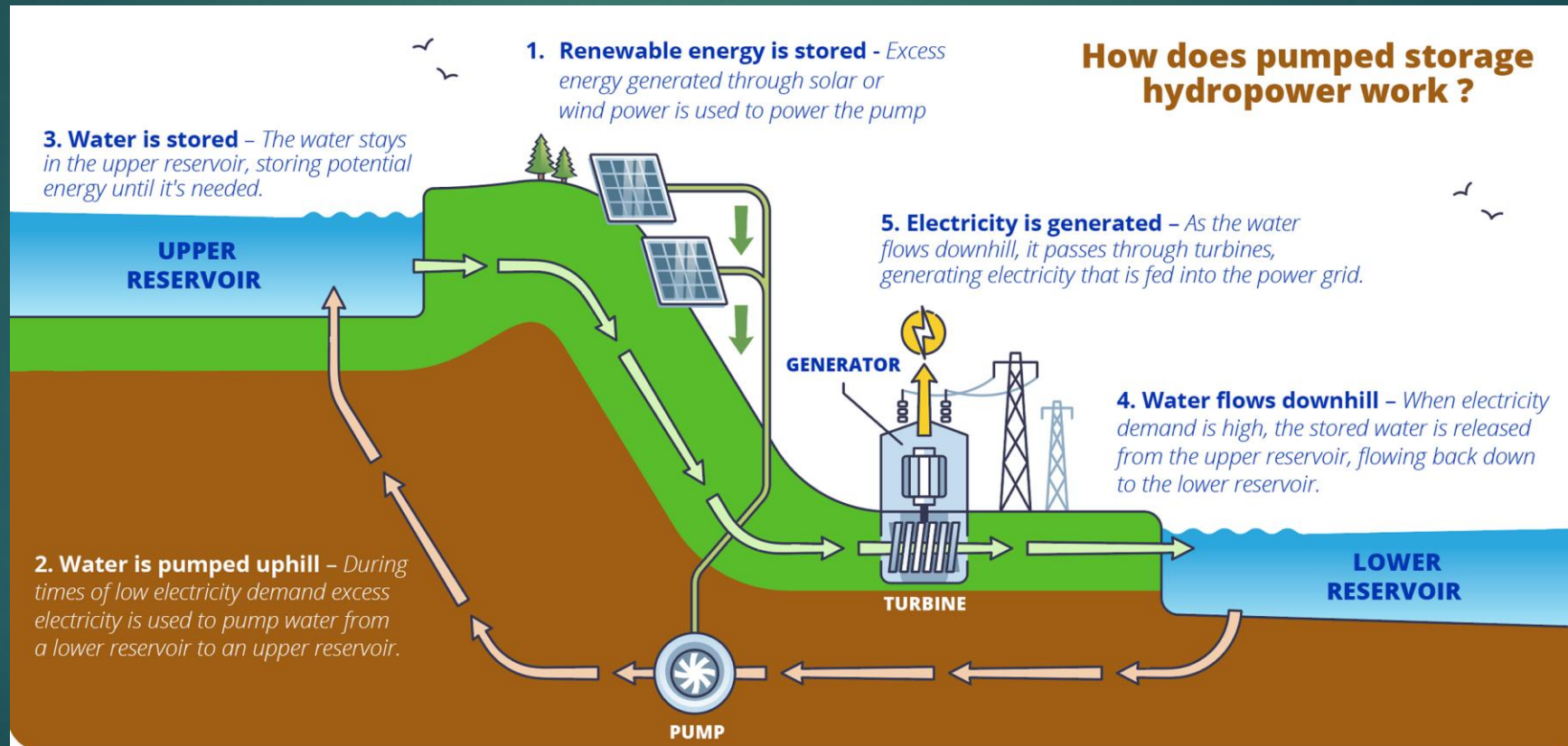
Impoundment (uses a dam and reservoir)



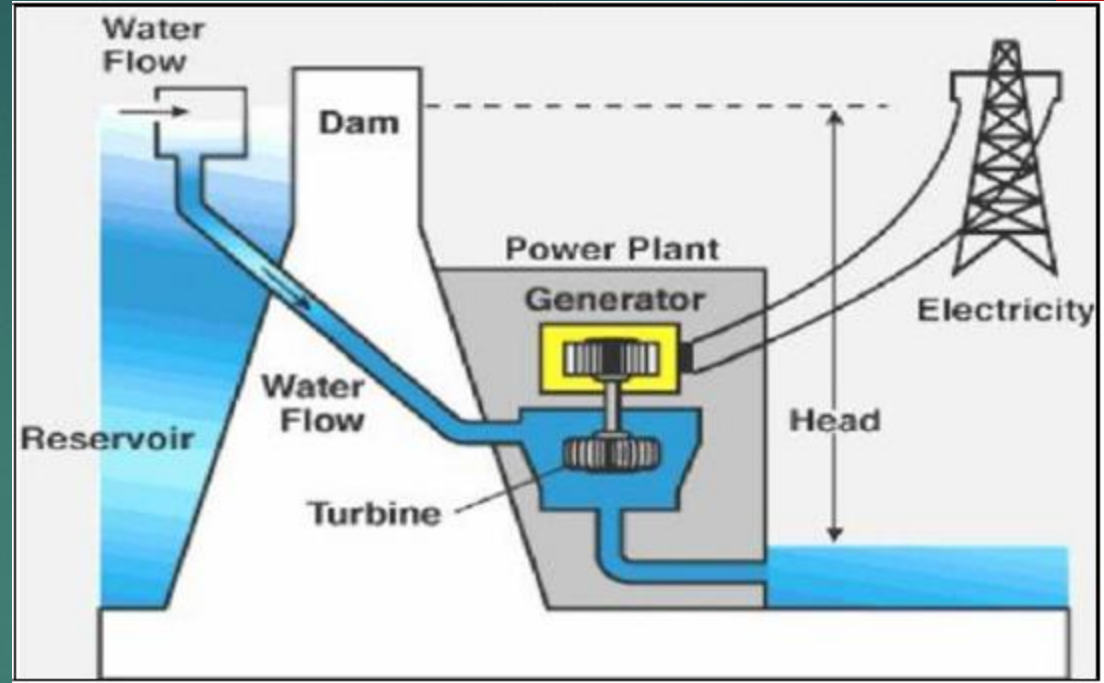
Diversion (uses a weir and channel to divert river flow)



Pumped Storage (uses two reservoirs to store and release water for energy storage)



$$P_h = \rho Q g H$$



$\rho \rightarrow$ Density of the fluid handled $\left(\frac{kg}{m^3}\right)$

$g \rightarrow$ Gravitational constant $\left(\frac{m}{s^2}\right)$

$H \rightarrow$ head (m)

$Q \rightarrow$ Flow rate $\left(\frac{m^3}{s}\right)$

Turbine

- ❖ Turbines are defined as the machines which convert the hydraulic energy into mechanical energy.
- ❖ This mechanical energy is used in running an electric generator which is directly coupled to the shaft of the turbine.
- ❖ Thus the mechanical energy is converted into electrical energy.
- ❖ The electric power which is obtained from hydraulic energy is known as *Hydro-electric power*.

Classification of turbine

According to the type of energy available at inlet

(a) Impulse turbine (b) Reaction turbine

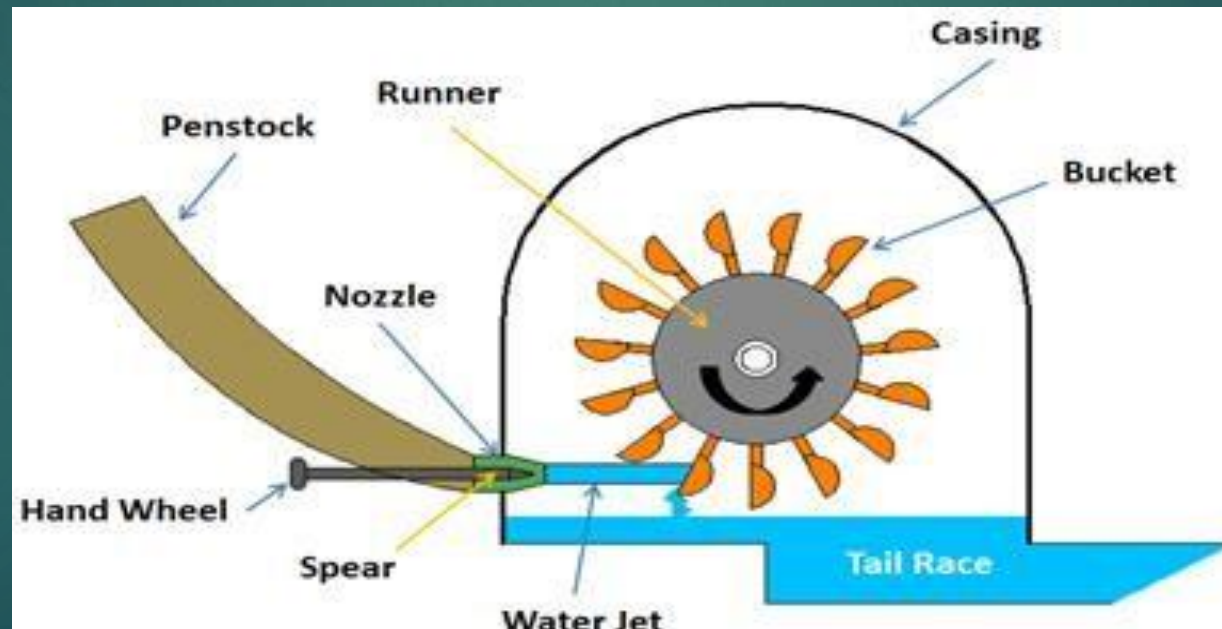
- ❖ If at the inlet of the turbine , **only kinetic energy**, the turbine is known as ***impulse turbine***.
 - e.g. Pelton Turbine.

- ❖ If at the inlet of the turbine, water possesses **kinetic energy** as well as **pressure energy**, the turbine is known as ***reaction turbine***.
 - e.g. (a) Francis turbine, (b) Kaplan turbine.

Impulse Turbine

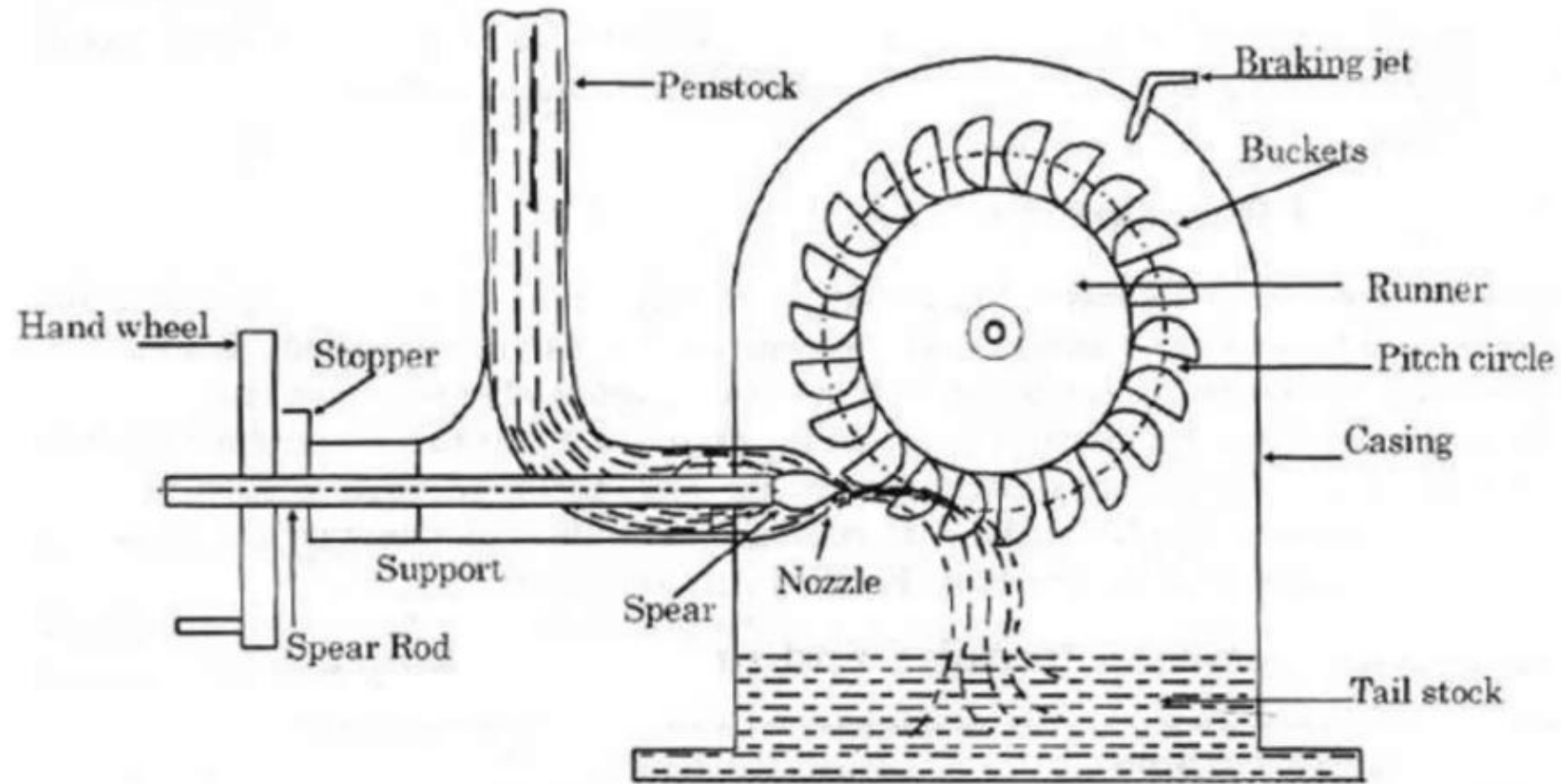
Main parts of Impulse turbine are :-

1. Nozzle and flow regulating device
2. Runner and bucket
3. casing



Impulse Turbine

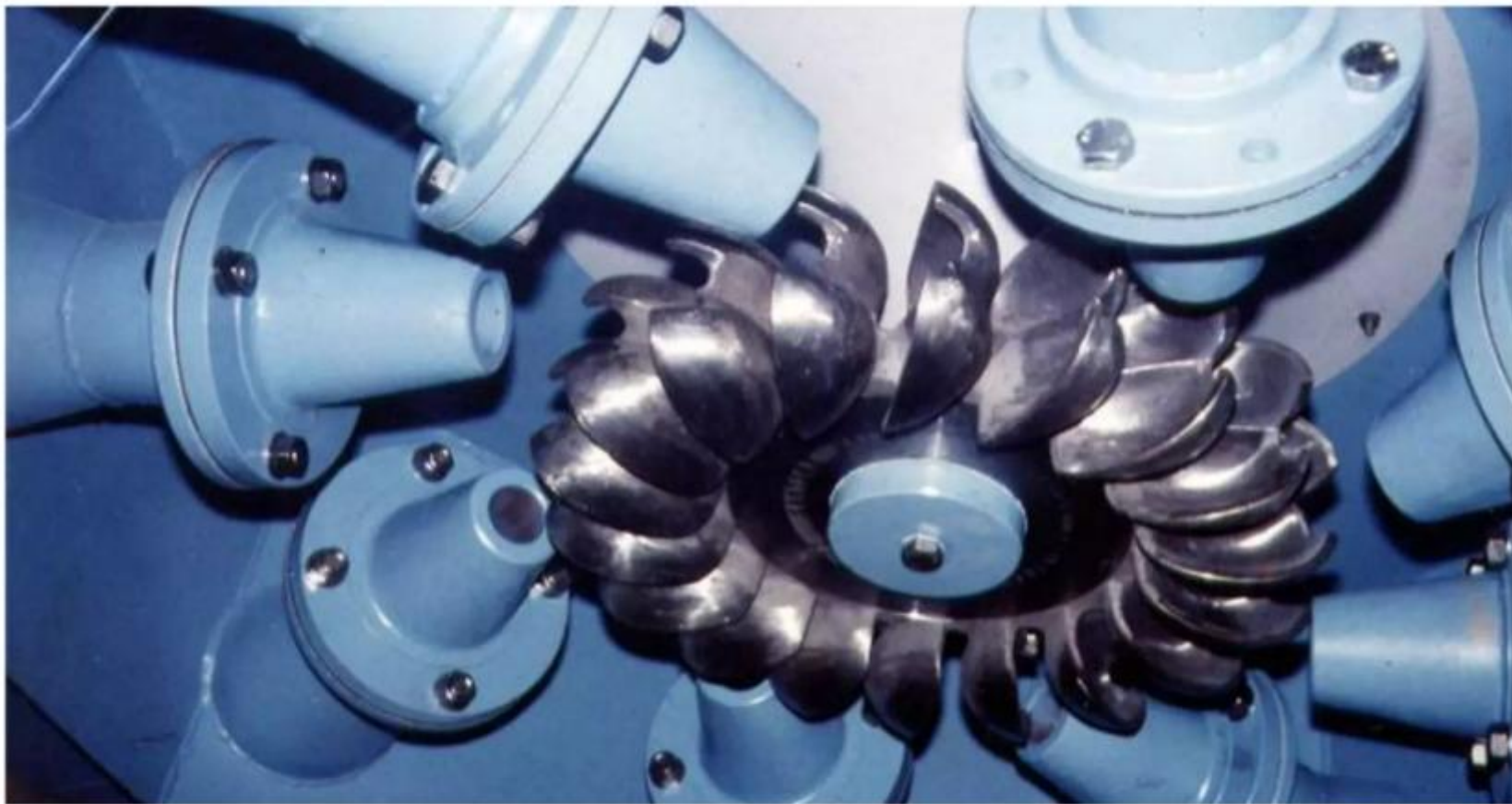
- ❖ **Nozzle with guide mechanism-** It is provided to convert the pressure energy into kinetic energy in the form of jet and it also regulates the quantity of water according to the load on turbine.
- ❖ **Runner-** A wheel of the turbine consist of series of buckets/blades/vanes mounted on its periphery.
- ❖ **Casing-** It is used to avoid accident and prevents the splashing of water. It does not perform any hydraulic function. The pressure throughout the turbine from inlet to outlet is **atmospheric** in case Impulse turbine.



PELTON WHEEL



PELTON



PELTON WHEEL WITH MULTILE JETS

Governing of Turbines

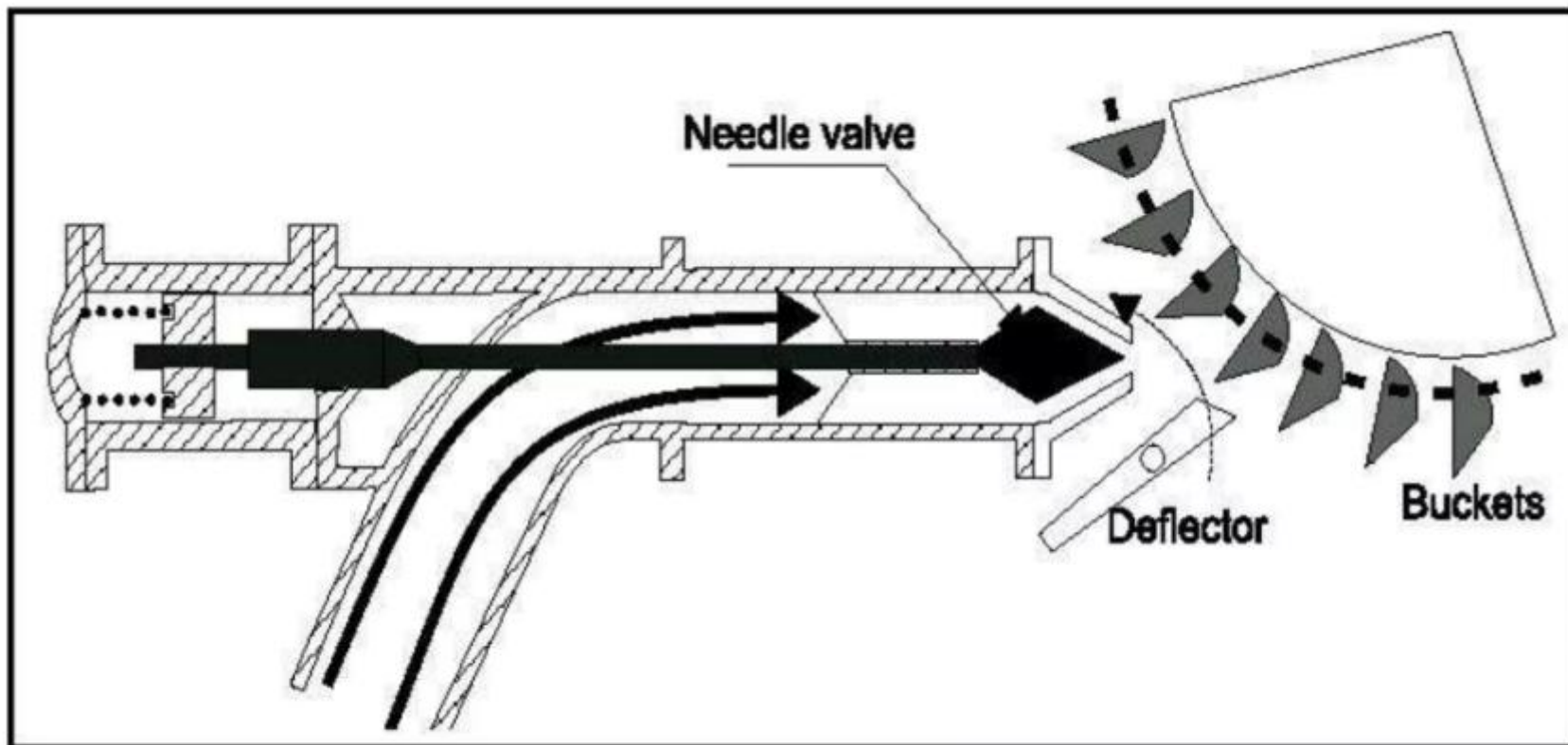
Governing means Speed Regulation.

Governing system or governor is the main controller of the hydraulic turbine. The governor varies the water flow through the turbine to control its speed or power output.

1. Impulse Turbine

- a) Spear Regulation
- b) Deflector Regulation
- c) Combined

2. Reaction Turbine

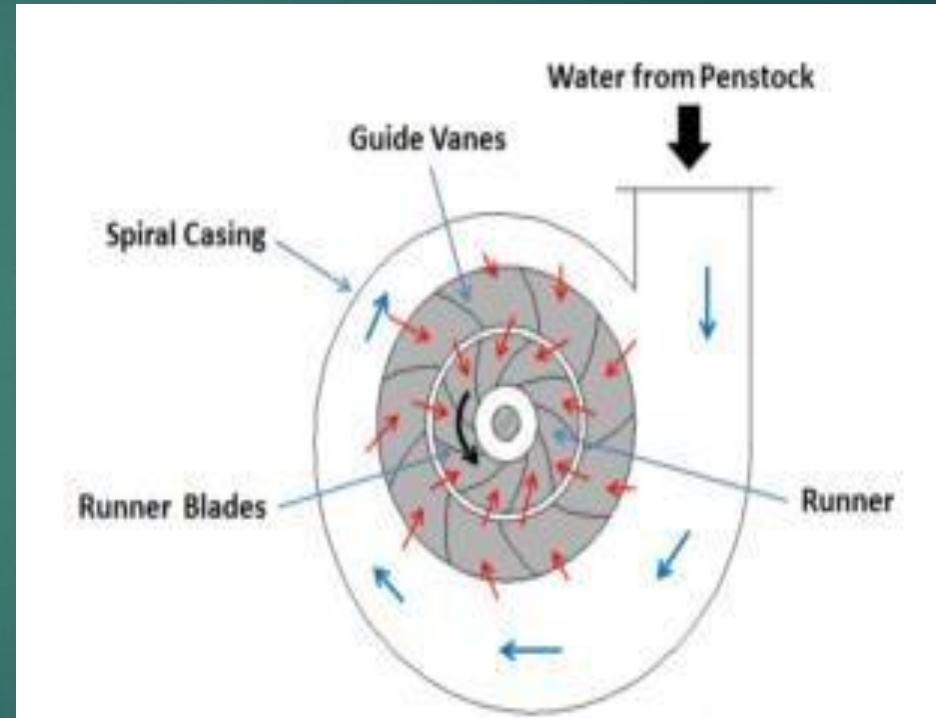


Governor of Pelton Wheel

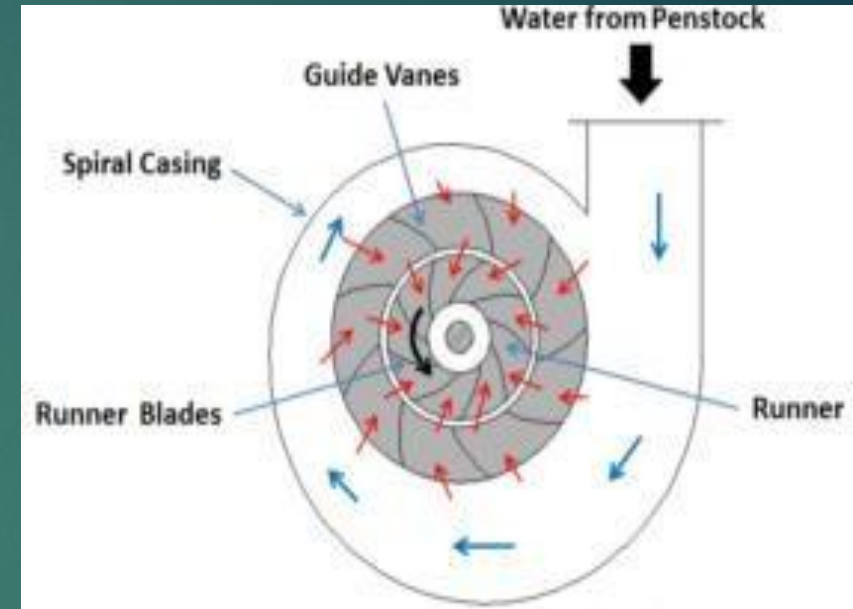
Reaction Turbine

Main parts of reaction turbine are :-

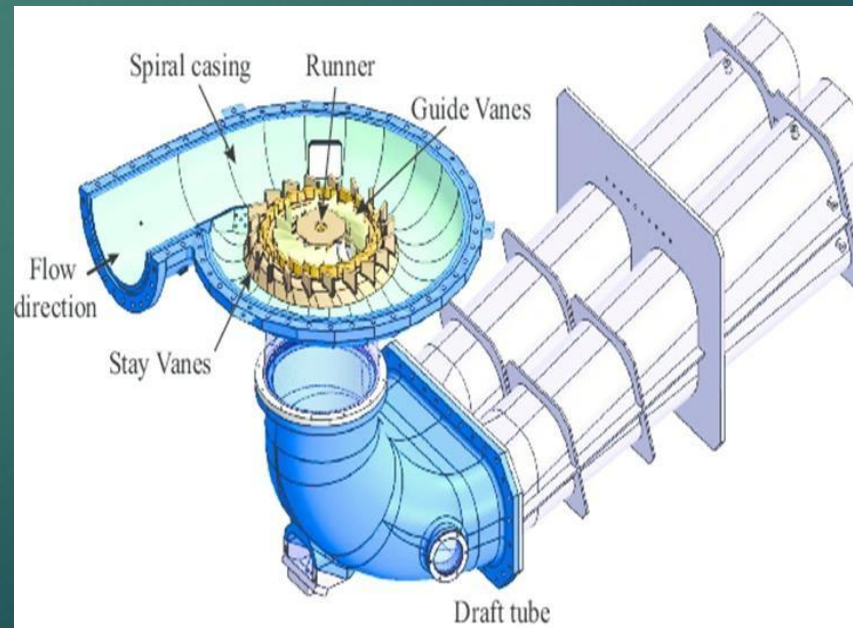
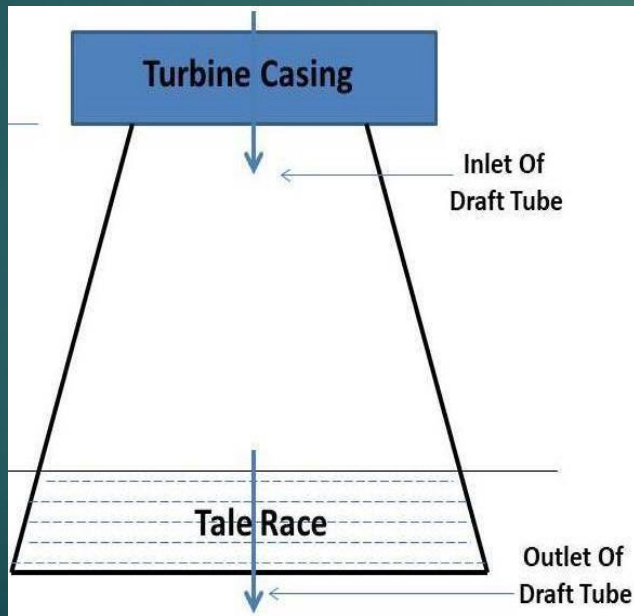
1. Casing
2. Guide mechanism
3. Runner
4. Draft tube



- ❖ **Casing-** In reaction turbine, casing and runner are always full of water. It is of spiral shape.
- ❖ **Runner-** A wheel of the turbine consist of series of buckets/blades/vanes mounted on its periphery.
- ❖ **Guide mechanism-** The guide vanes allow the water to strike the fixed blades on the runner without shock at inlet.



- ❖ **Draft tube-** Draft Tube is a diverging tube fitted at the exit of runner of turbine and used to utilize the kinetic energy available with water at the exit of runner.
- ❖ Pressure head is increased by decreasing the exit velocity.
- ❖ Overall efficiency and the output of the turbine can be improved.



Comparison of IMPULSE and REACTION turbine

• Impulse Turbine

- $HE \rightarrow KE \rightarrow ME$
- Principal of Impulse
- Wheel revolves in open air
- No pressure difference at I/L and O/L of runner
- Casing has not to perform any hydraulic function
Casing has to prevent splashing of water
- $P_1 = P_2 = P_{atm}$ $V_1 \gg V_2$
- Pelton, Girard, Turgo, Banki

• Reaction Turbine

- $HE \rightarrow KE+PE \rightarrow ME$
- Principal of Impulse-reaction
- Wheel submerge in water
- Pressure difference at I/L and O/L of runner
- Involute Casing has to perform hydraulic function i.e to produce moment on the runner
- $P_1 > P_2$ $V_1 > V_2$
- Francis, Kaplan, Propeller

• Impulse Turbine

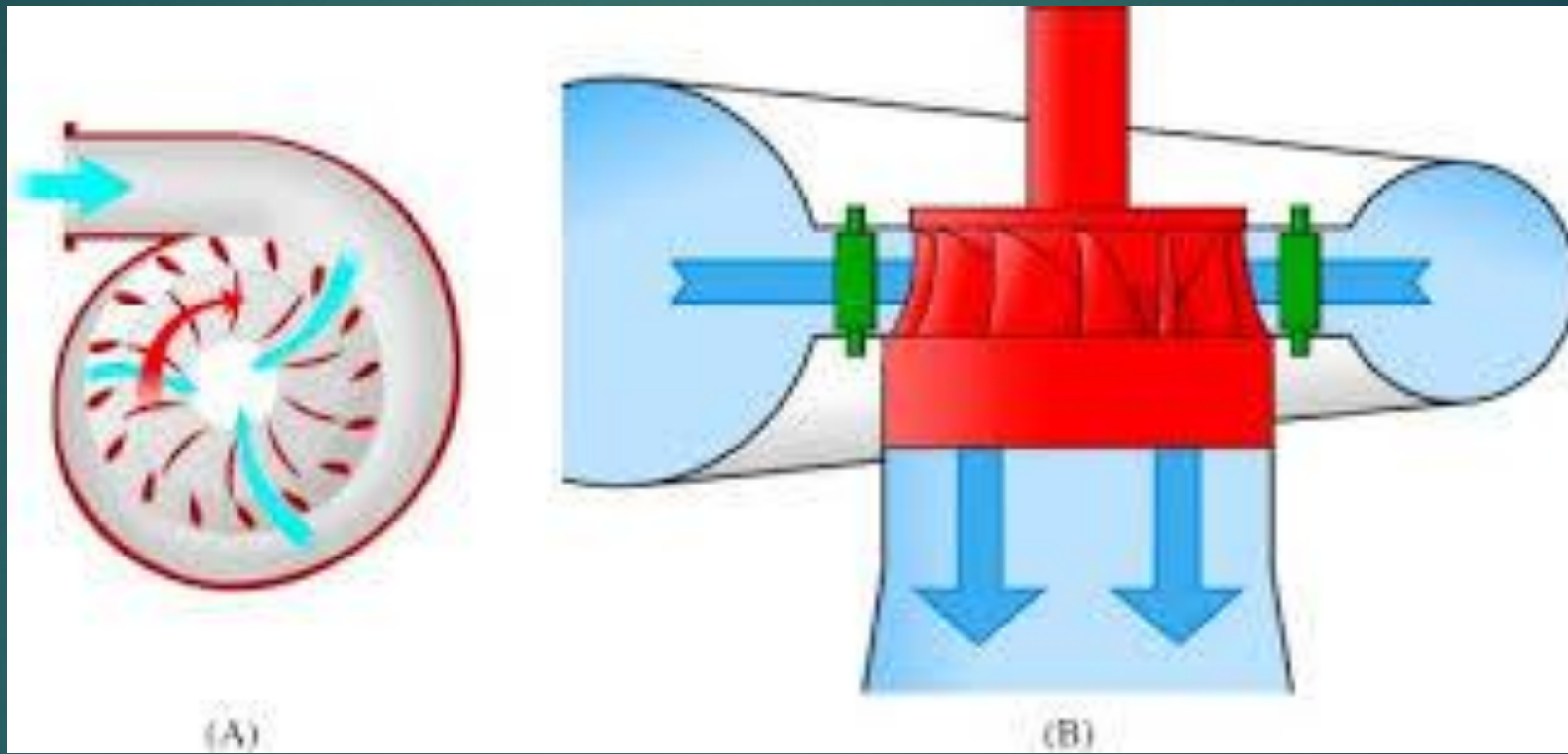
- Used for high head
- Energy transfer takes place due to impulsive action
- Casing need not be air tight
- No cavitation
- For same head and power size of turbine is large
- Simple in construction but poor efficiency

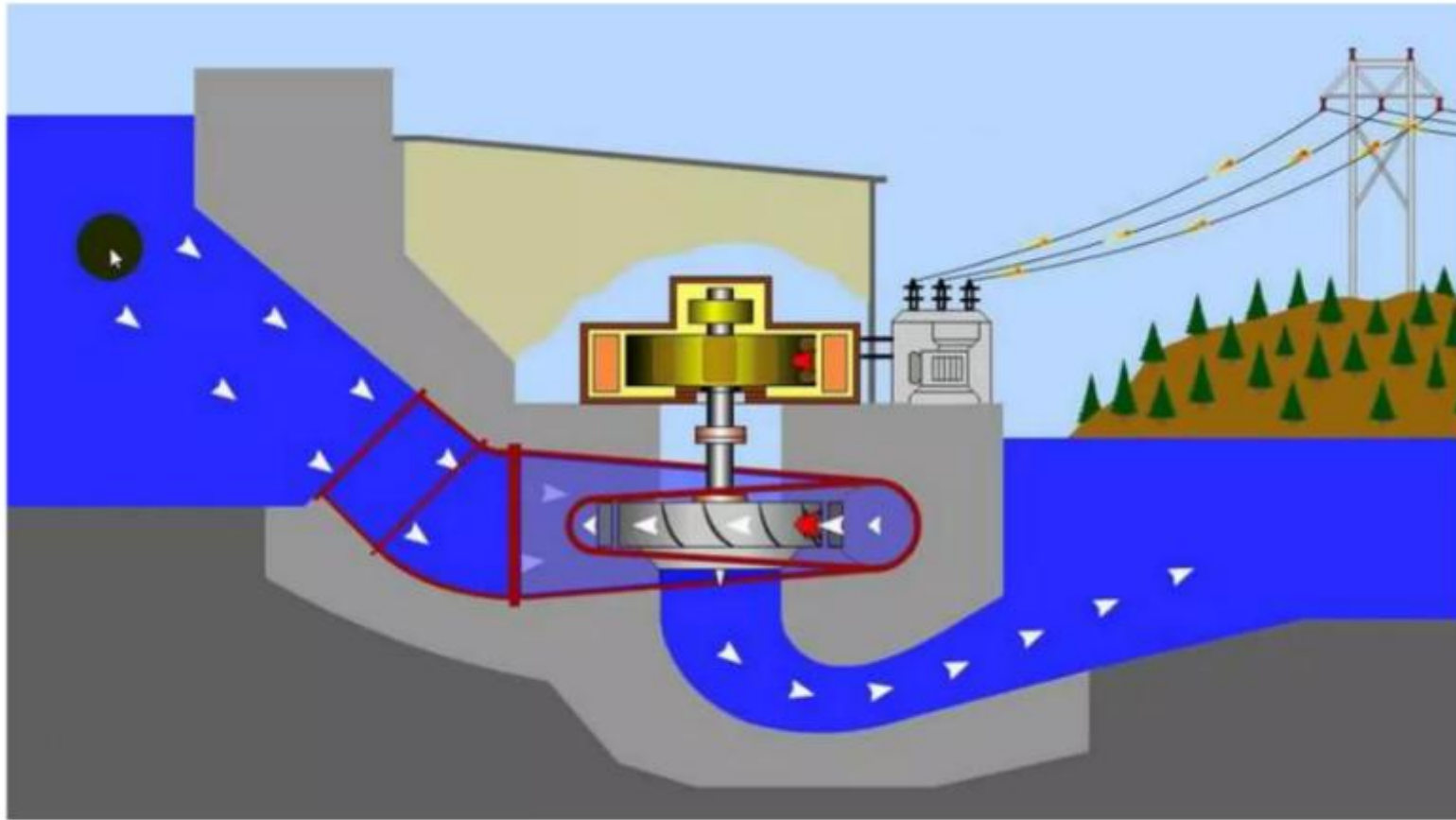
• Reaction Turbine

- Used for low head
- Energy transfer takes place due to reaction between flowing water and vane surface & impulsive action
- Casing must be air tight
- Cavitation occur
- Relatively small
- Complicated in construction but relatively high efficiency

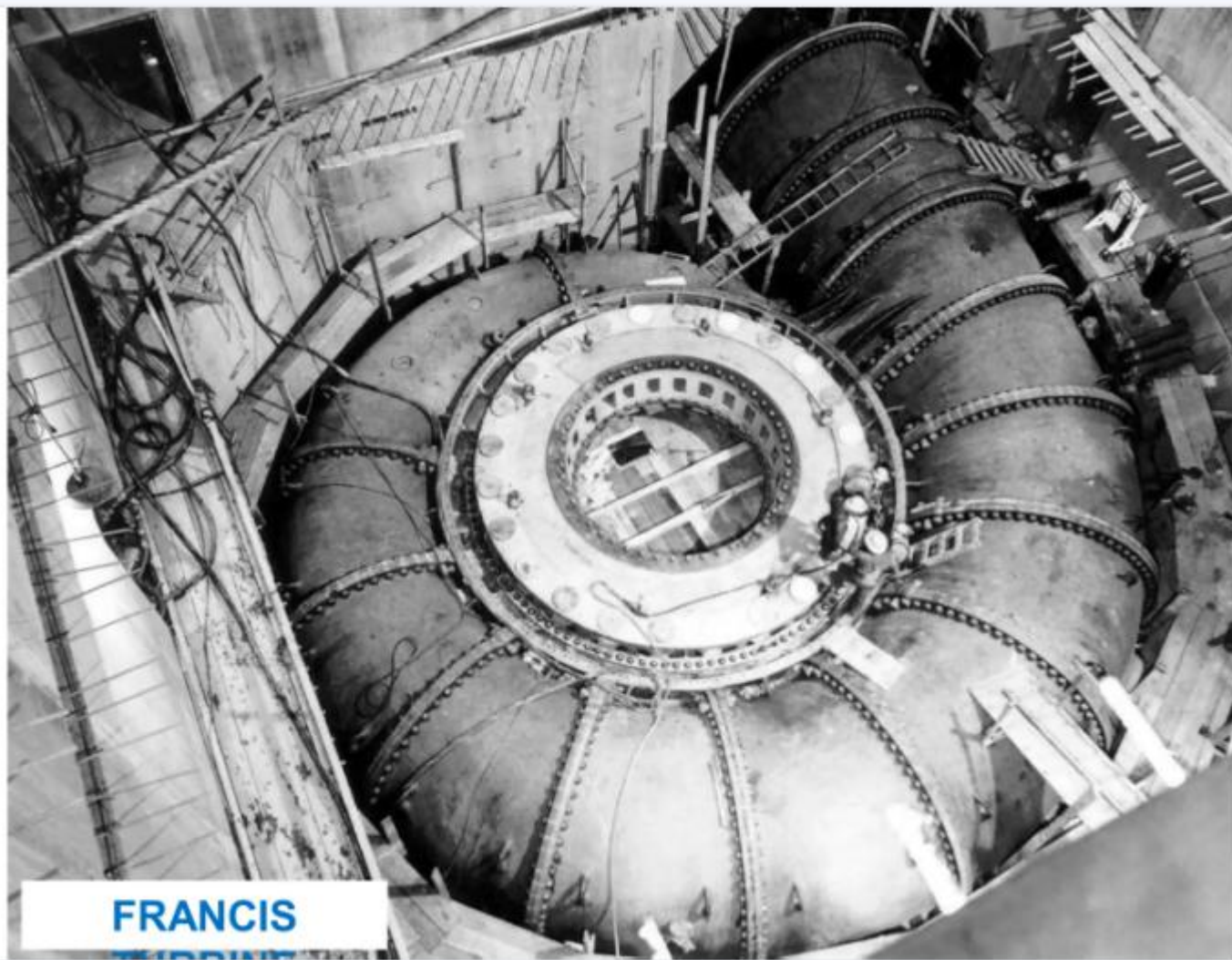
(a) Francis Turbine

- Francis Turbine is the first hydraulic turbine with **radial inflow**. If the water flows radially through the runner, from outwards to inwards then it is known as an inward radial flow turbine.
- Francis turbine is a reaction turbine as the energy available at the inlet of the turbine is a combination of **kinetic and pressure energy**.





FRANCIS TURBINE



FRANCIS

Guide Vanes



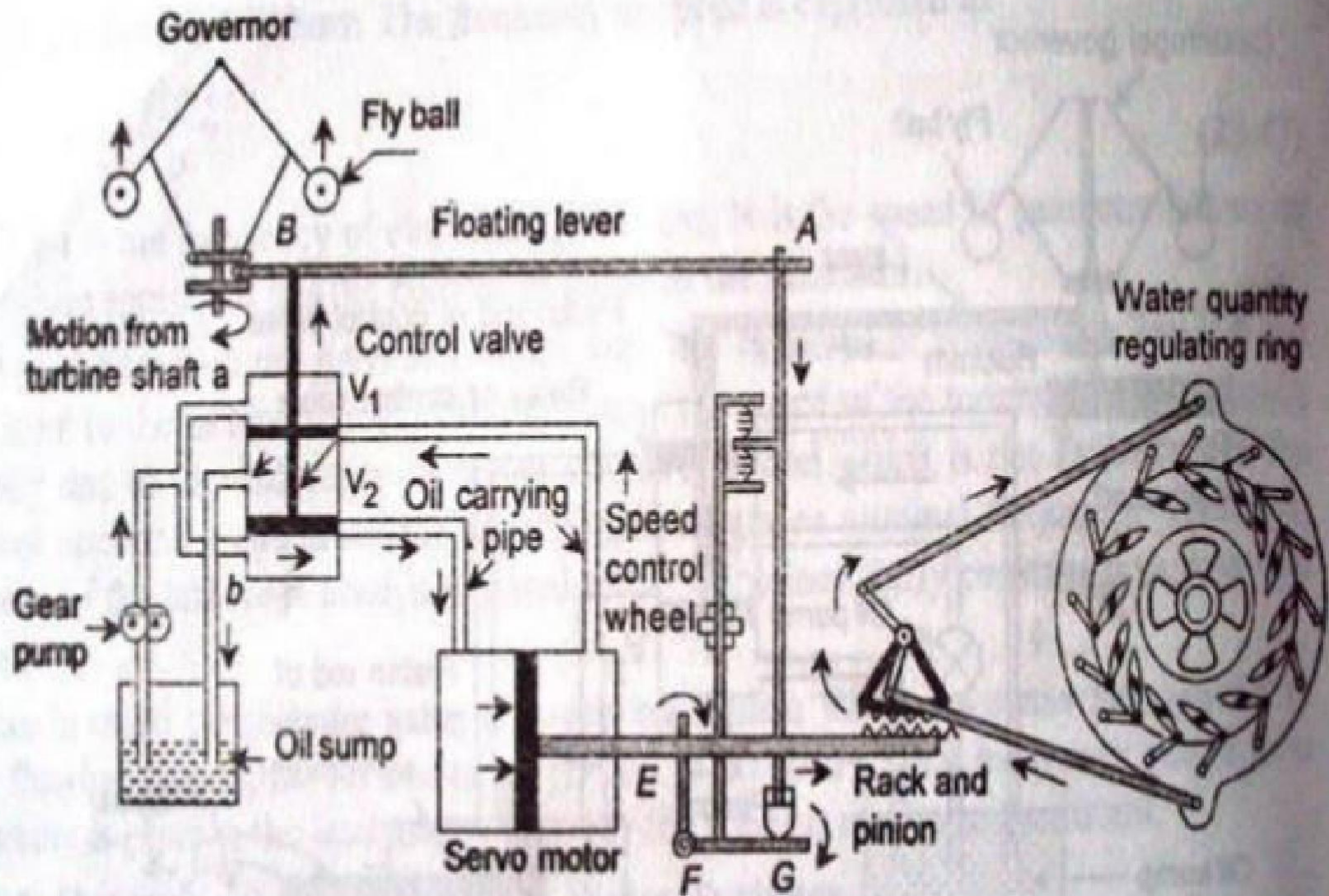


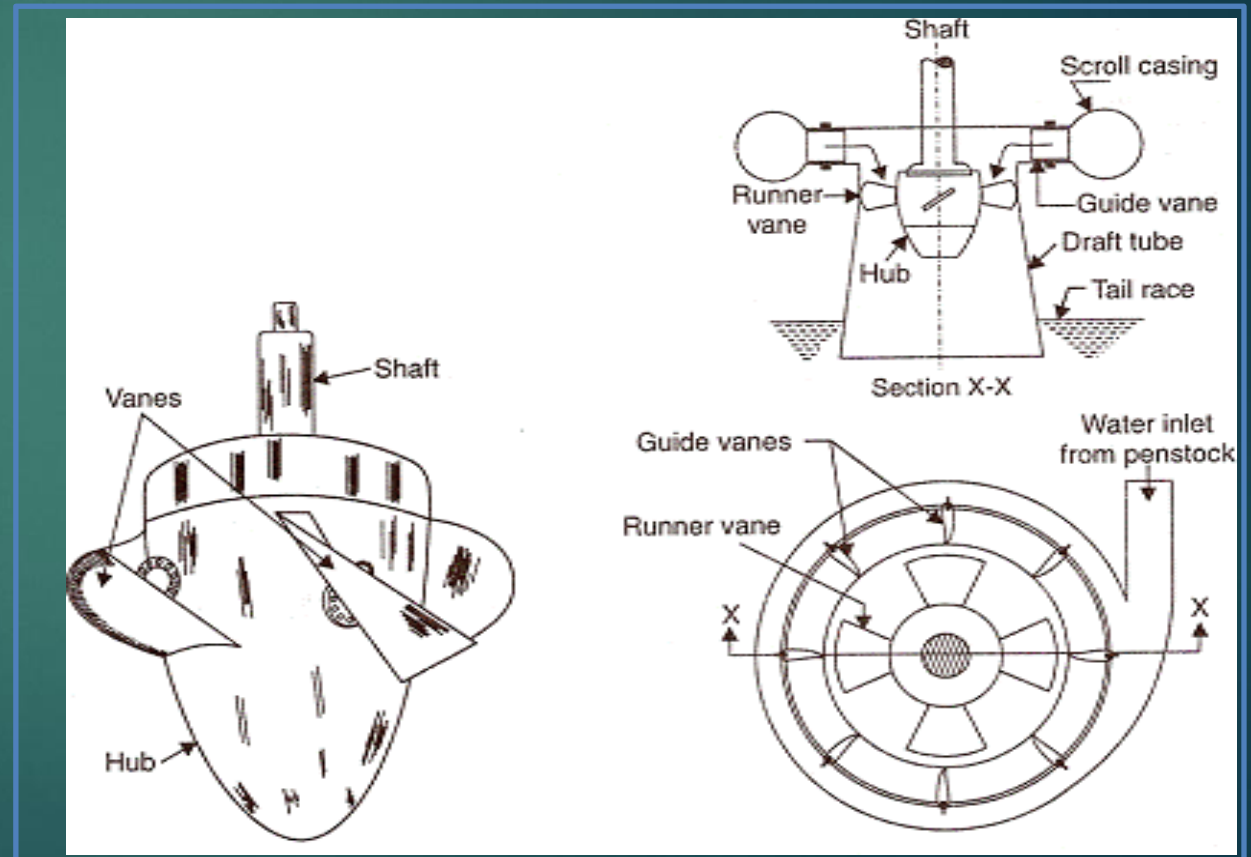
Fig. (1) Governing of Francis Water Turbine

(b) Kaplan turbine

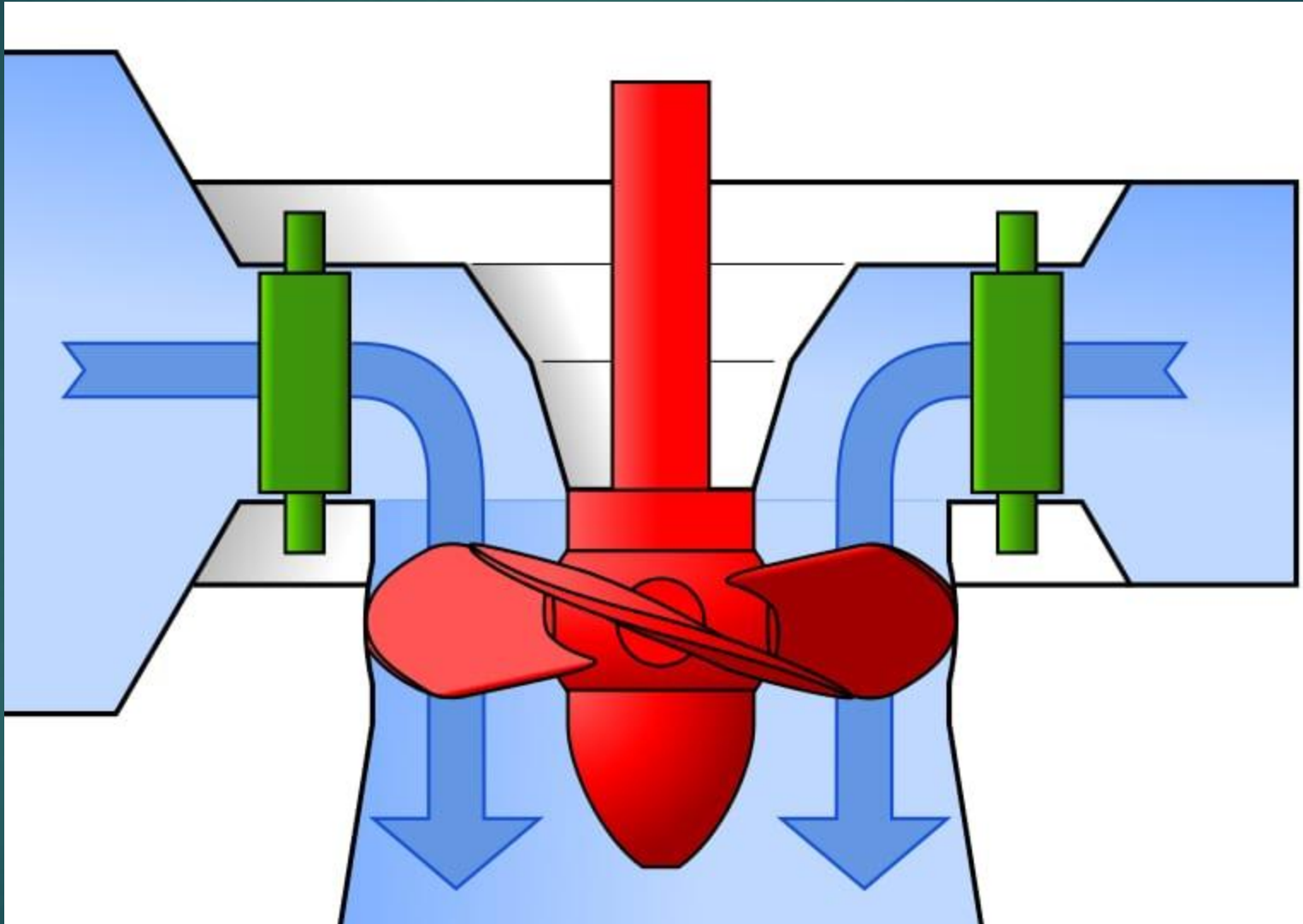
- Kaplan turbine is an **axial flow reaction turbine**. The water flows through the runner of the turbine in an axial direction and the energy at the inlet of the turbine is the sum of **kinetic and pressure energy**.
- In an axial flow reaction turbine the **shaft is vertical**. The lower end of the shaft is larger and is known as '*hub*' or '*boss*'. It is on this hub that the vanes are attached.
- If the **vanes are adjustable** then it is known as *kaplan Turbine* and if the vanes are **non adjustable** then it is known as *Propeller Turbine*.
- Kaplan turbines widely used throughout the world in **high-flow, low-head power production**.

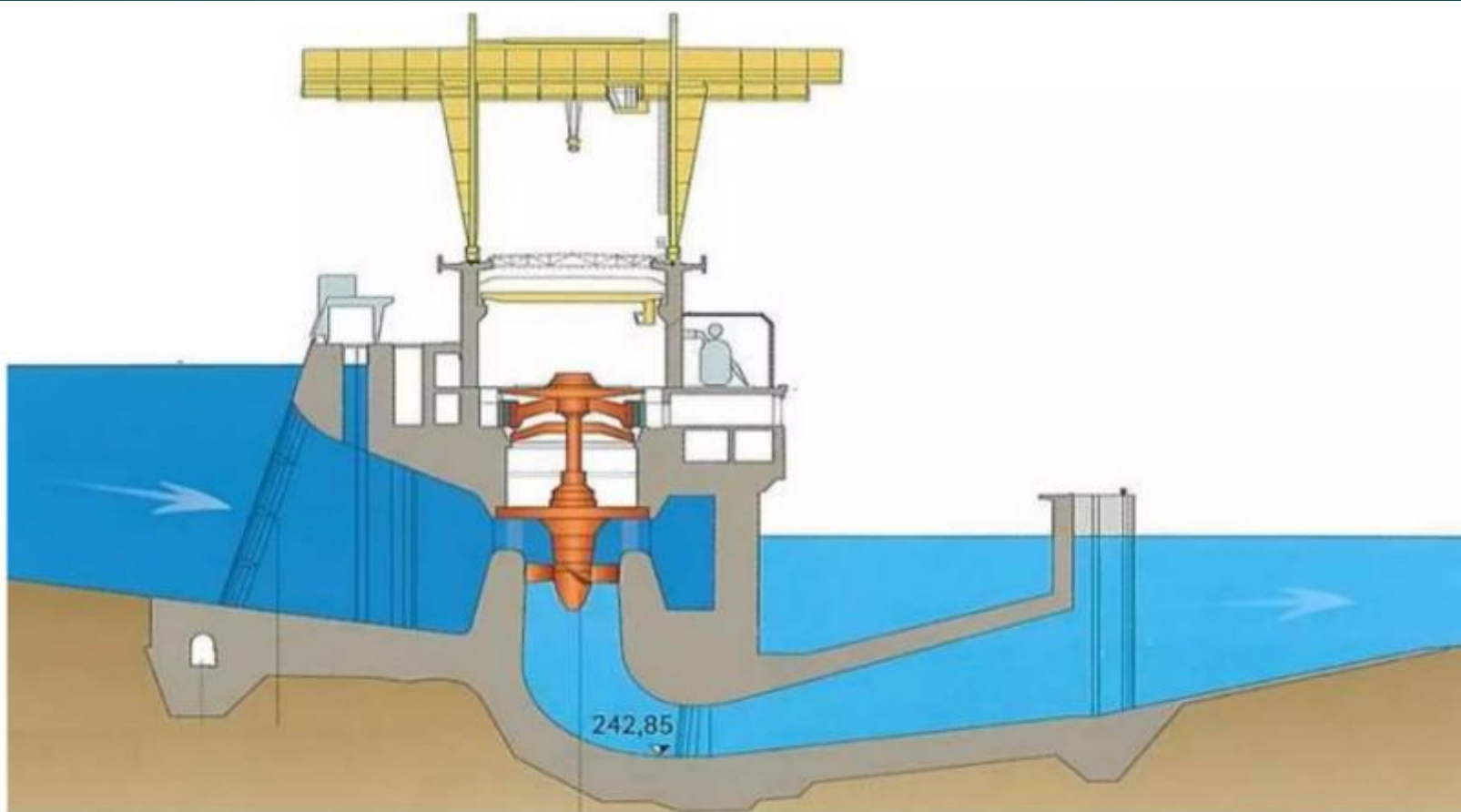
•The main parts of a **kaplan Turbine** are:

1. Scroll Casing
2. Guide vane Mechanism
3. Hub with Vanes
4. Draft Tube



Kaplan





KAPLAN TURBINE

Comparison of the Turbines

	Pelton	Francis	Kaplan
Energy at inlet	Impulse	Reaction	Reaction
Head	High	Medium	Low
Flow Direction	Tangential	Mixed (Radial entry and axial outlet)	Axial flow
Discharge Required	Low	Medium	High
Specific Speed	Low	Medium	High

Ocean Energy

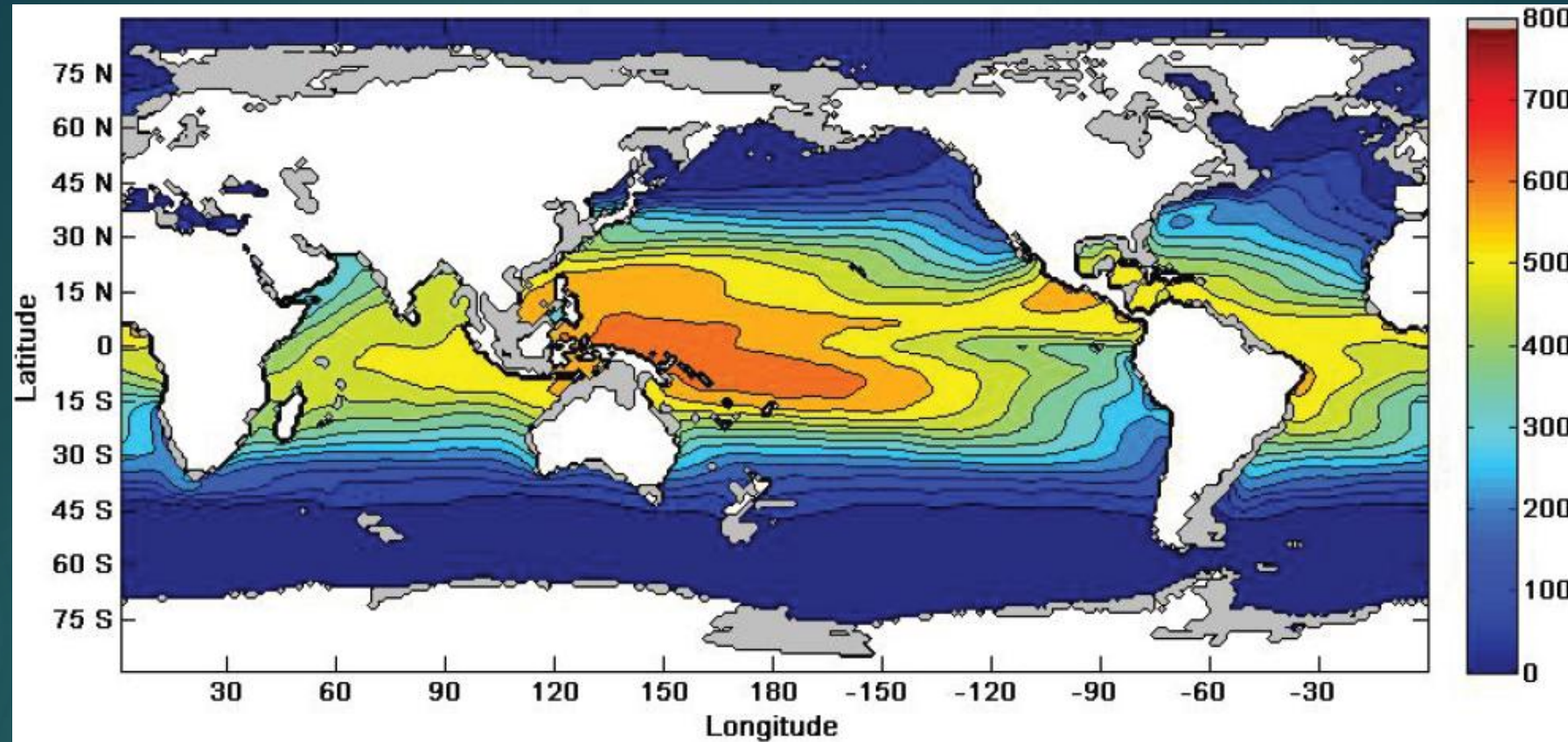
INTRODUCTION

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- A source of renewable energy, uses the heat energy stored in the earth's oceans to generate electricity.
- Works best when the temperature difference between warmer and the colder water is at least 20°C
- Exists in tropical coastal areas roughly between the tropic of Capricorn and the tropic of cancer.



Global map of the annual OTEC power density available in kW/(m³/s)



High Potential : Western Pacific Ocean (near the equator)

Medium Potential : Central & Eastern Pacific Ocean (towards South America)

Central Atlantic Ocean (West Africa, Caribbean region), Indian Ocean (Equatorial belt)

Low Potential : High latitudes (North Atlantic, North Pacific, Southern Ocean near Antarctica)

Current scenario of Ocean Thermal Energy Conversion (OTEC) system

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► Current Status:

- Only a few small OTEC plants are running, in places like Japan and Hawaii.
- New projects are being tested, mostly near tropical regions where the ocean is warm.
- OTEC can also help make fresh water from seawater.

► Challenges:

- OTEC needs a big temperature difference (about 20°C) to work well.
- It is expensive and works best in certain regions.

Pilot projects:

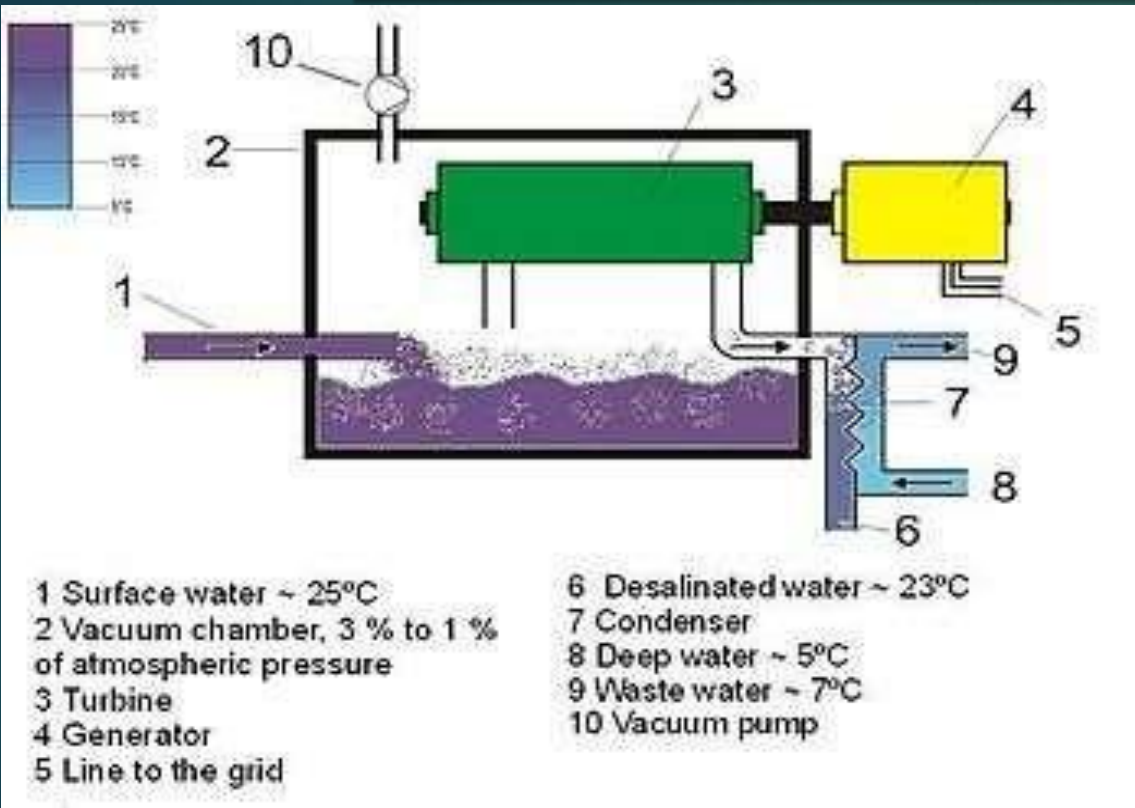
- ▶ Hawaii (USA): A 100 kW closed-cycle plant **operated** by Makai Ocean Engineering at the Natural Energy Laboratory of Hawaii Authority (NELHA) connected to the grid since 2015.
- ▶ Japan: Saga University **operates** a 50 kW plant in Okinawa for research, with plans to upgrade to a 1 MW facility.
- ▶ India: In 2024, the National Institute of Ocean Technology (NIOT) announced **plans** for a 165 kW floating OTEC and desalination plant near the Lakshadweep Islands.
- ▶ South Korea: The Korea Research Institute of Ships and Ocean Engineering (KRISO) **operates** a floating 1 MW OTEC plant.

open-cycle or closed-cycle

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- ▶ OTEC (Ocean Thermal Energy Conversion) systems can be **open-cycle or closed-cycle**, differing in their working fluid:
- ▶ **open-cycle systems** use warm seawater to produce steam for a turbine, while **closed-cycle systems** use a low-boiling-point fluid like ammonia that circulates through heat exchangers to power the turbine.
- ▶ **Open-cycle** OTEC offers the benefit of producing fresh water as a byproduct, while **closed-cycle systems** avoid the risk of working fluid leaks and are more efficient in certain configurations.

OTEC TECHNOLOGIES (Open cycle)

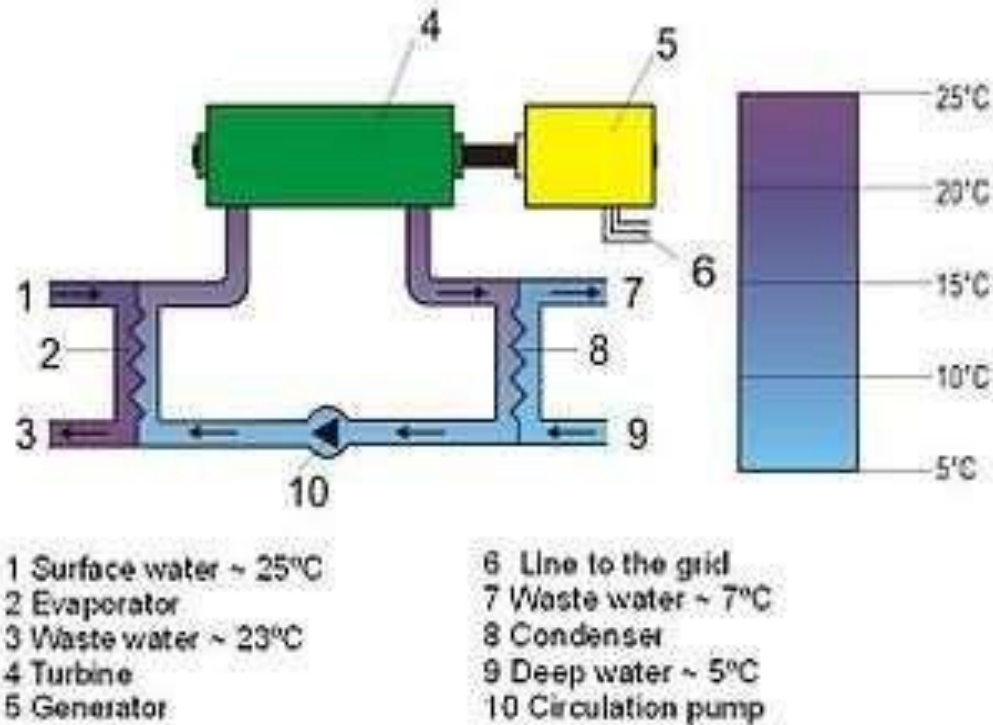


Working Fluid: Warm, surface seawater.

Process: Warm seawater is pumped into a low-pressure chamber, where it boils and turns into steam. This steam then drives a turbine and a generator. The steam is then condensed using cold deep-sea water and the water can be collected as fresh water.

Key Feature: Produces freshwater as a byproduct.

OTEC TECHNOLOGIES (Closed cycle)



Working Fluid: A working fluid with a low boiling point, such as ammonia or propane.

Process: Warm surface seawater is used in a heat exchanger to vaporize the working fluid. The resulting vapor turns a turbine and generator. Cold deep-sea water is then used to condense the vapor, which is pumped back to the evaporator to restart the cycle.

Key Feature: The working fluid remains in a closed loop, circulating continuously.

ENVIRONMENTAL IMPACT

Potential Environmental Benefits:

- **Zero emissions and clean energy:** OTEC generates renewable energy without greenhouse gas emissions or pollutants, making it a very clean alternative to fossil fuels.
- **Climate change mitigation:** Widespread OTEC implementation can reduce sea surface temperatures by up to 3.1 °C, suppress ocean acidification, and reduce atmospheric CO₂ concentrations by displacing fossil-fuel-generated electricity.
- **Continuous and scalable power:** OTEC can provide a stable, firm power source that is scalable and versatile, suitable for tropical regions.

Potential Environmental Concerns

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- ▶ **Marine Ecosystem Disruption:** Large OTEC facilities withdraw and discharge massive volumes of water, which can change local temperature and nutrient balances, affecting the marine food web and local ecosystems.
- ▶ **Entrainment and Impingement:** Marine organisms (like fish eggs, larvae, and plankton) can get trapped against intake screens or pulled into the OTEC system, harming some species.
- ▶ **Chemical and Physical Changes:** Upwelling of deep water is often more acidic and lower in oxygen, which might impact surface species and change ecosystem dynamics.
- ▶ **Localized and Polar Warming:** While OTEC may cool tropical seas, some models suggest it could slightly warm polar regions over very long periods.
- ▶ **Construction and Noise:** Building onshore or nearshore OTEC plants can disturb wildlife, degrade air quality, and change coastal habitats during construction.

Challenges of Ocean Thermal Energy Conversion (OTEC)

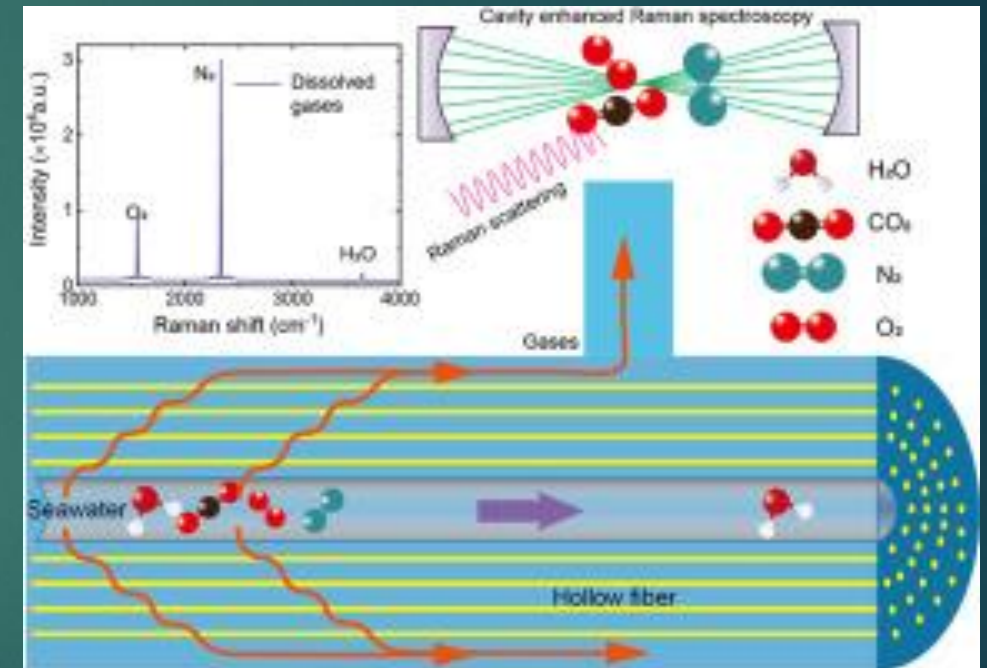
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- ▶ Low Efficiency:
 - ▶ OTEC converts heat to energy but with low efficiency (~4-7%) because ocean temperature difference is small (about 20°C).
- ▶ High Cost:
 - ▶ Building and running OTEC plants, especially offshore, is expensive with high initial investment.
- ▶ Engineering Difficulties:
 - ▶ Handling large volumes of warm and cold seawater is complex.
 - ▶ Dealing with dissolved gases in seawater requires special equipment.
- ▶ Limited Suitable Locations:
 - ▶ OTEC works best in tropical oceans where warm surface and cold deep water are present.
- ▶ Environmental Concerns:
 - ▶ Changing ocean temperature and nutrient flows can affect marine ecosystems and wildlife.

Dealing with dissolved gases in seawater requires special equipment.

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- In an open-cycle OTEC system, a seawater deaerator is used to remove dissolved gases from the warm seawater before it enters the flash evaporator.
- This deaerator often employs porous aeration stones to create small bubbles that rise through the seawater, drawing out gases like nitrogen and oxygen.
- The extracted gases are then collected in a low-pressure chamber and withdrawn by a vacuum compressor.



ADVANTAGES

- ✓ Totally Renewable.
- ✓ Generates electricity with no green house effect.
- ✓ Produces desalinated water.

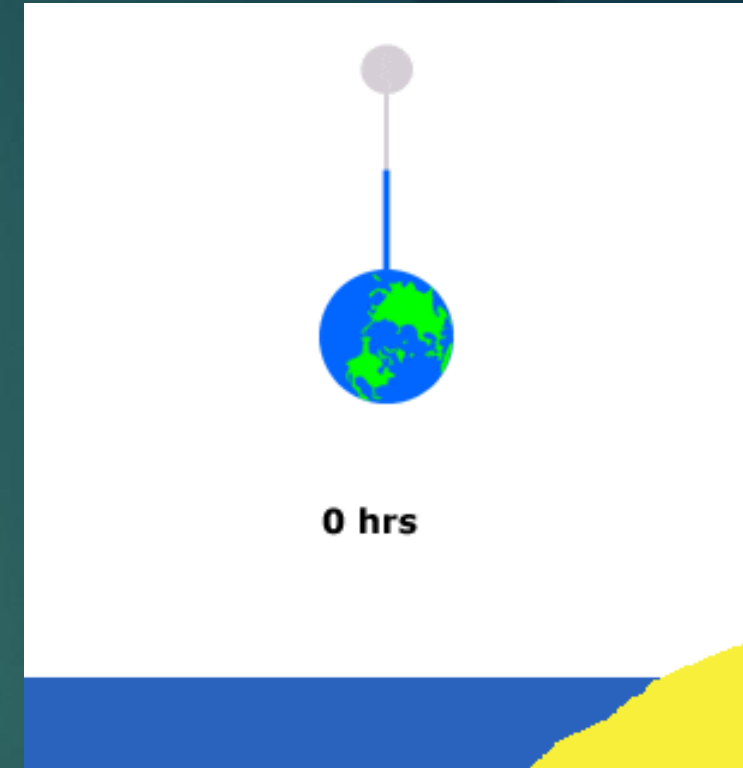
DISADVANTAGES

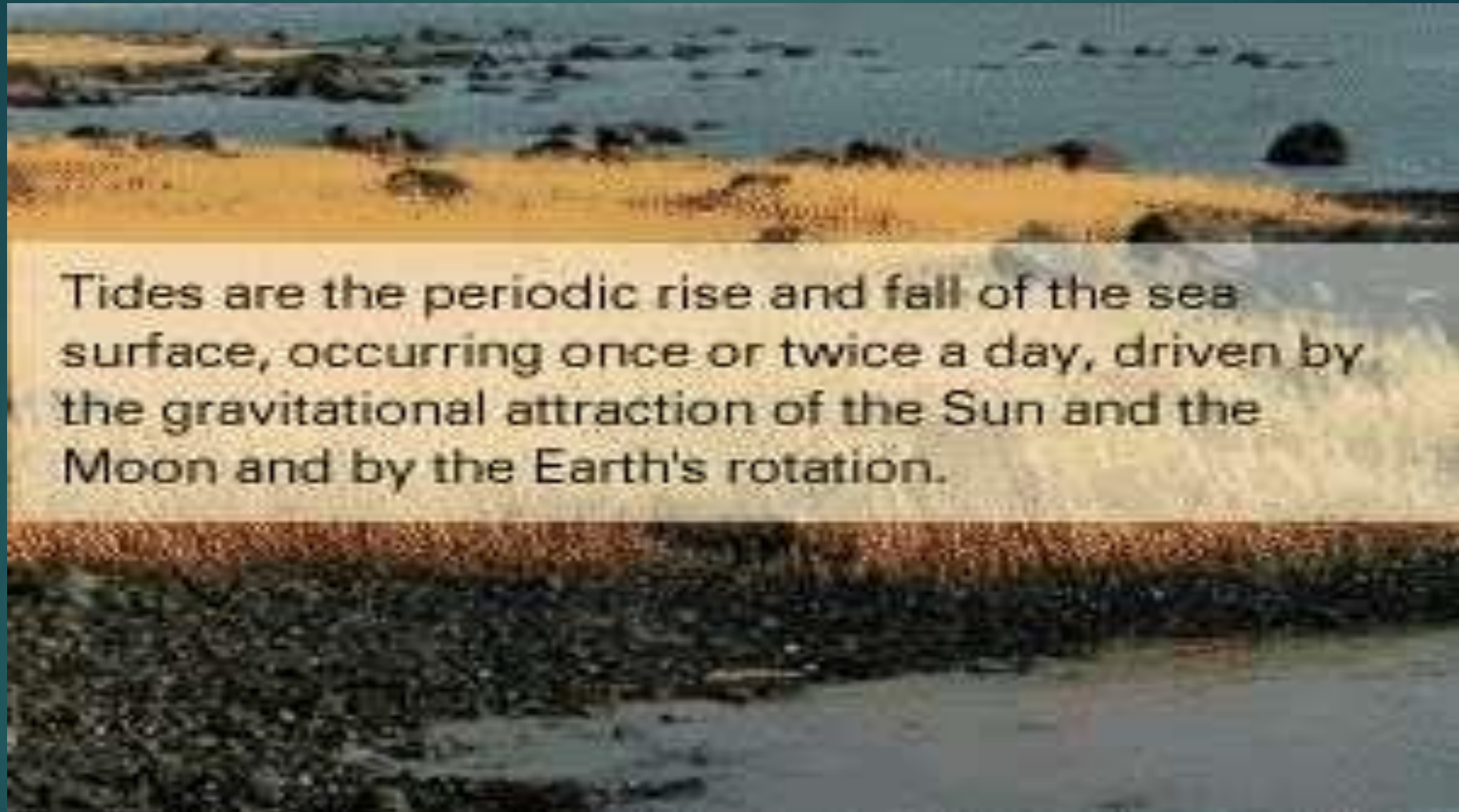
- ✓ Needs a large difference in temperatures for best results.
- ✓ Needs to be close to the national grid.
- ✓ Plant needs safe location from storms and surf.

Tidal energy - single basin and double basin plants. Wave energy conversion systems,

INTRODUCTION

- Tidal power, also called **TIDAL ENERGY**, is a form of **HYDROPOWER** which converts the energy of tides into the useful form of power, mainly in electricity.
- Tides are the waves caused due to **gravitational pull** of the moon and sun.
- Ocean tides are the **periodic rise and fall of ocean water level** occurs **twice in each lunar day**.
- During one lunar day the ocean water level rises twice and fall twice.
- **Time interval** between a **consecutive low tide and high tide** is **6.207 hrs**.
- Tidal range is the difference between the consecutive high tide and low tide.
- During high tide, the water flow into the dam and during low tide water flow out which result in moving the turbine.
- Tides are more predictable than wind energy and solar power.

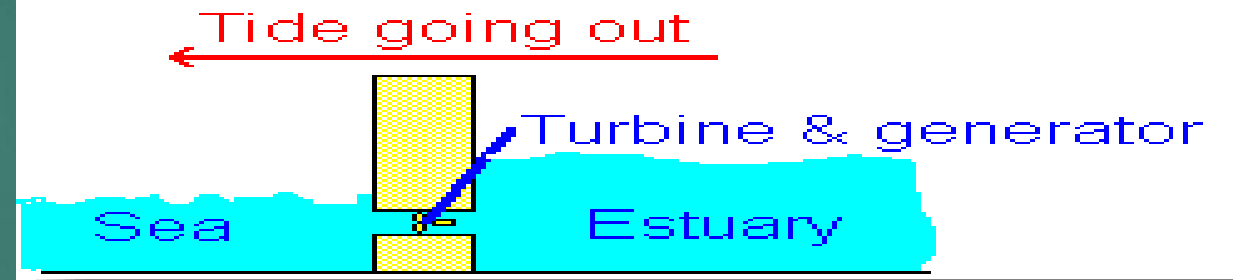
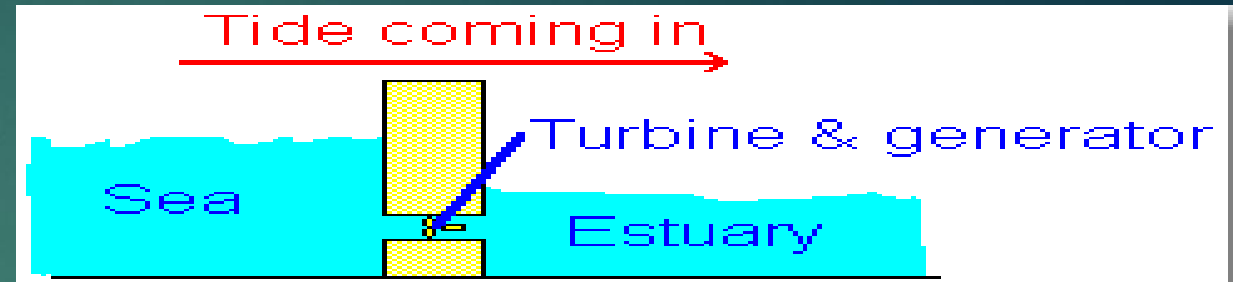




Tides are the periodic rise and fall of the sea surface, occurring once or twice a day, driven by the gravitational attraction of the Sun and the Moon and by the Earth's rotation.

TIDES:-

- ❖ This tidal electricity generation works as the tides comes in and again when it goes out.
- ❖ The turbine is driven by the power of the sea in both direction.



Types of tides



Main parts of TPP:-

- A tidal power plant consists of **three main parts**:
- The first being the **barrage** itself, holding the water back during high tide.
- The second part is the **sluice gate** that let water through the third part,
- The third part consist **turbine and generator**, resulting in electricity generation.

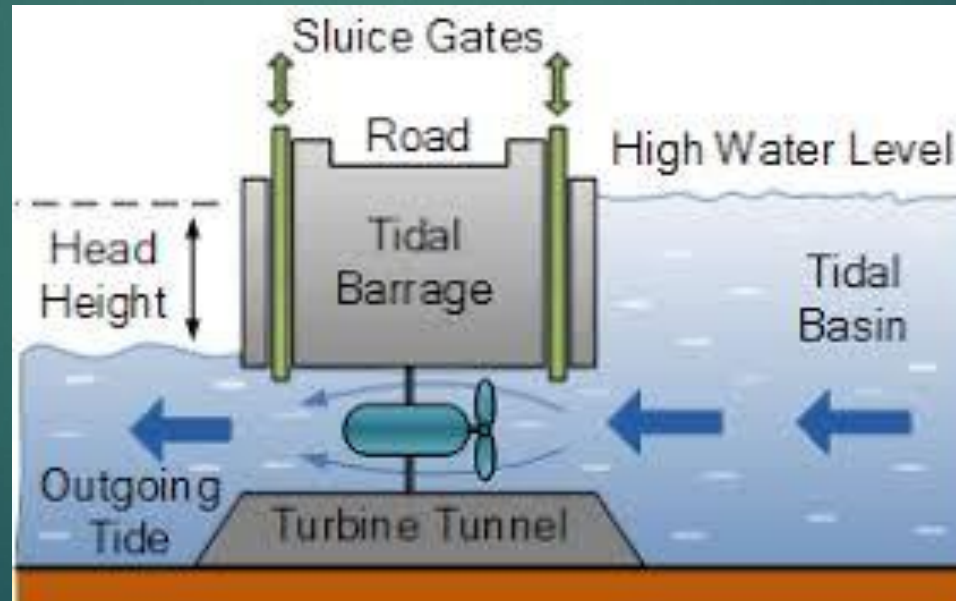
1. Tidal Barrage

- A **tidal barrage** is a dam-like structure used to capture the energy from masses of water moving in and out of a bay or river due to tidal forces.



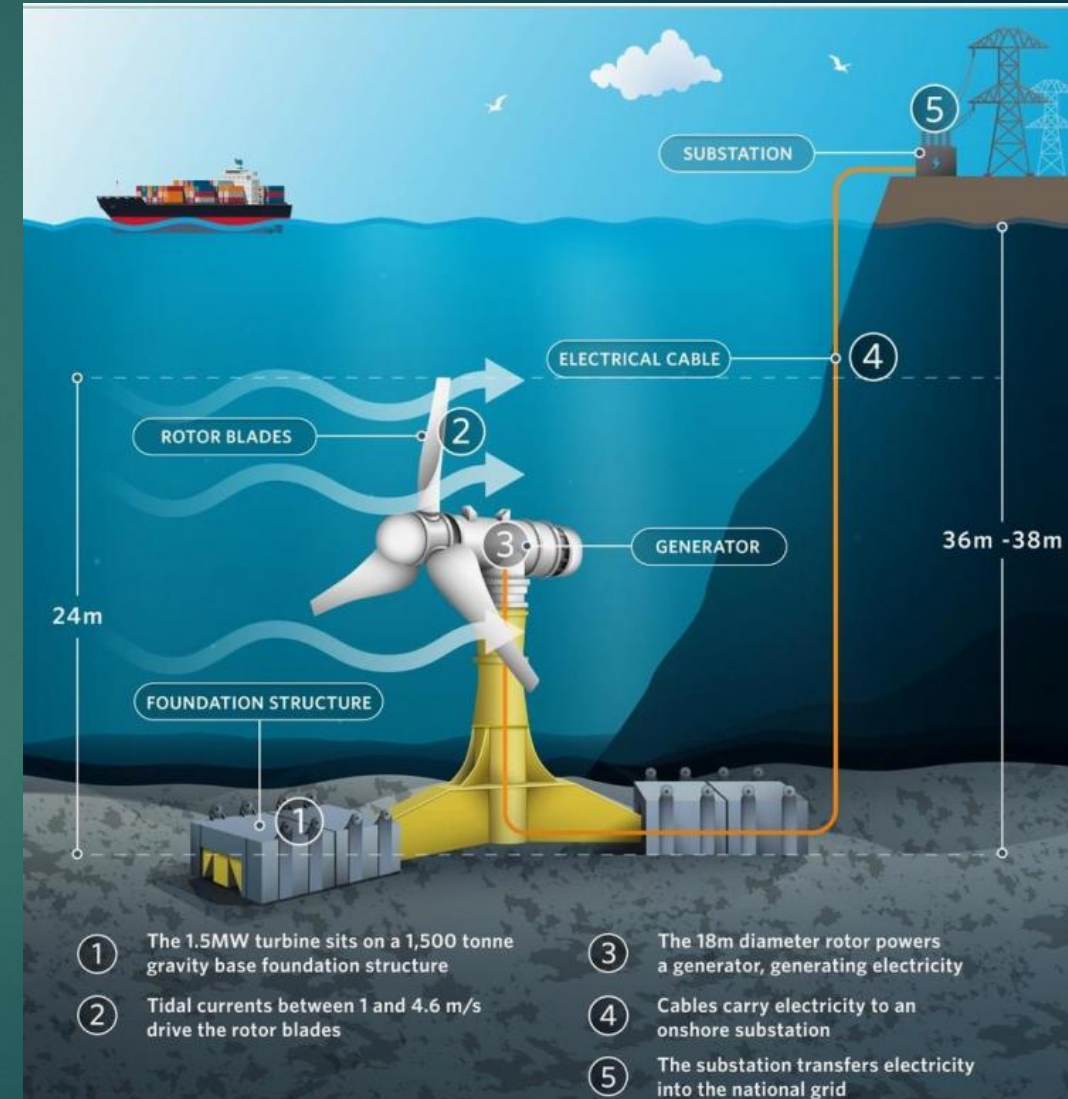
2. Sluice Gate

- The sluice gates are left **open during high tide** and **closed during low tide** to create a water level differential, creating a potential difference that powers the turbine when the water is released.



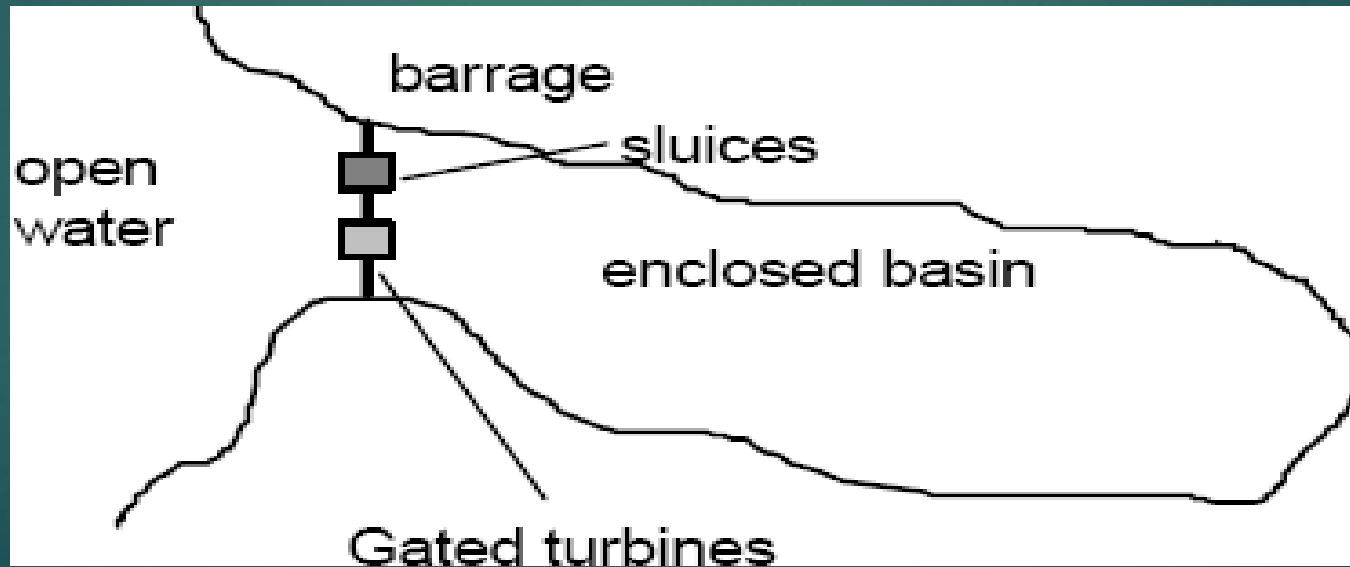
3. Tidal Stream Generator

- Tidal stream generators are very similar to wind turbines except they are below the water surface instead of above or on land.
- The turbine and generator convert the movement of water coming from change in tide, the kinetic energy, into electricity.
- Water is 830 times denser than air and therefore can generate electricity at lower speeds than wind turbines.



Single Basin Scheme:

This scheme has one barrage and one water storage basin, one way system, the incoming tide is allowed to fill the basin through sluice ways during the tide and the impounded water is used to generate electricity by letting the water flow from basin to the sea through the turbines during single basin schemes is intermittent generation power.



Single-Basin Tidal Power Plant

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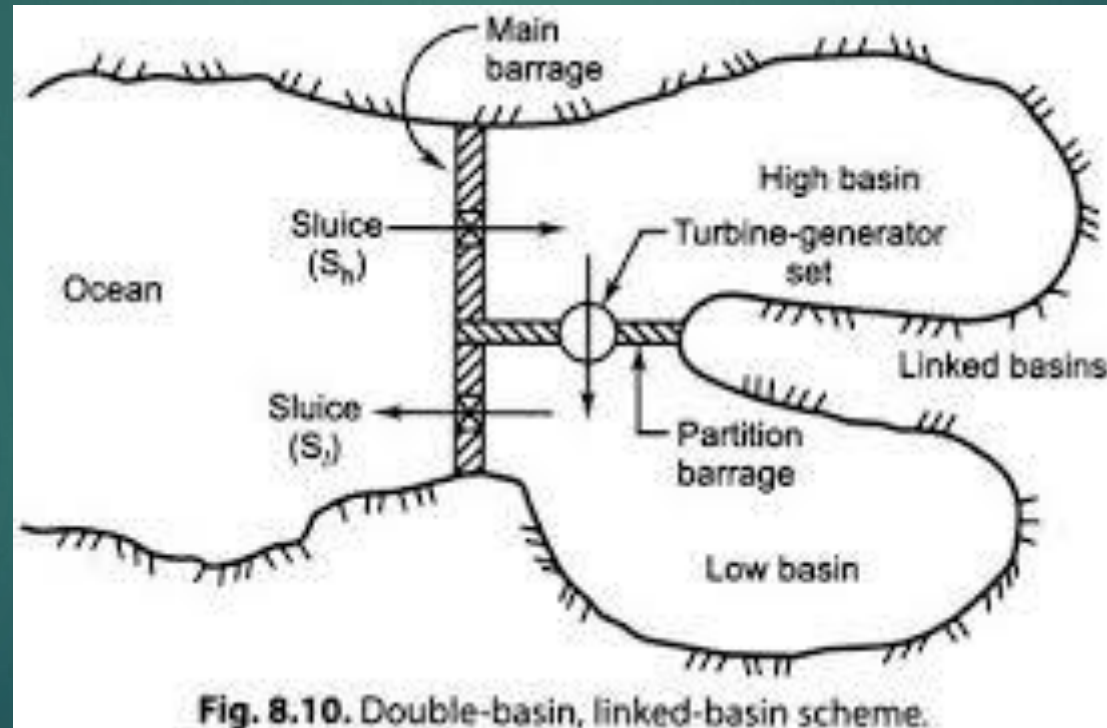
Structure: A single enclosed basin is built between the sea and the land, separated by a dam containing turbines and sluice gates.

Working Principle:

- During high tide, the basin fills with seawater through the sluice gates, storing water.
- As the tide goes out (ebb tide), the basin level is higher than the sea, creating a head (difference in water level).
- This stored water then flows through turbines to the sea, generating electricity.

Power Output: This system typically generates power only during the ebb tide. Power generation is intermittent and for a shorter duration compared to double-basin systems.

Double Basin Scheme: In the double basin scheme, there are two basins on the landward side with the powerhouse located at the interconnecting waterway between the two basins



Double-Basin Tidal Power Plant

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Structure: Two basins are constructed at different levels, with a dam and powerhouse located between them.

Working Principle:

- The upper basin fills during high tide.
- The lower basin empties during low tide.
- This setup creates a permanent head between the two basins, regardless of the tide's position relative to the sea.
- Water can then flow between the upper and lower basins through turbines to generate electricity continuously or on demand, both during high and low tides.

Power Output: The main advantage is its ability to provide continuous or on-demand power, making it more flexible than a single-basin plant.

Advantages of Tidal Energy

- It is an **inexhaustible** source of energy.
 - Tidal energy is **environment friendly** energy and doesn't produce greenhouse gases.
 - As 71% of Earth's surface is covered by water, there is **scope to generate this energy on large scale**.
 - **Efficiency of tidal power is far greater** as compared to coal, solar or wind energy. Its efficiency is around 80%.
-
- Tidal Energy doesn't require any kind of fuel to run.
 - The life of tidal energy power plant is very long.
 - The large density of water, almost 1000 times greater than in air, results in very **large amounts of energy to get out of the tidal currents even if the speed is low**.

Disadvantages of Tidal Energy

- **Cost** of construction of tidal power plant is high.
- There are **very few ideal locations** for construction of plant and they too are localized to coastal regions only.
- **Intensity of sea waves is unpredictable** and there can be damage to power generation units.
- Influences aquatic life adversely and can disrupt **migration of fish**.
- The **actual generation is for a short period of time**. The tides only happen twice a day so electricity can be produced only for that time.
- Usually the places where **tidal energy is produced** are far away from the places where it is **consumed**. This transmission is expensive and difficult.

Introduction & Resource Potential

Title: What is Wave Energy & Its Resource

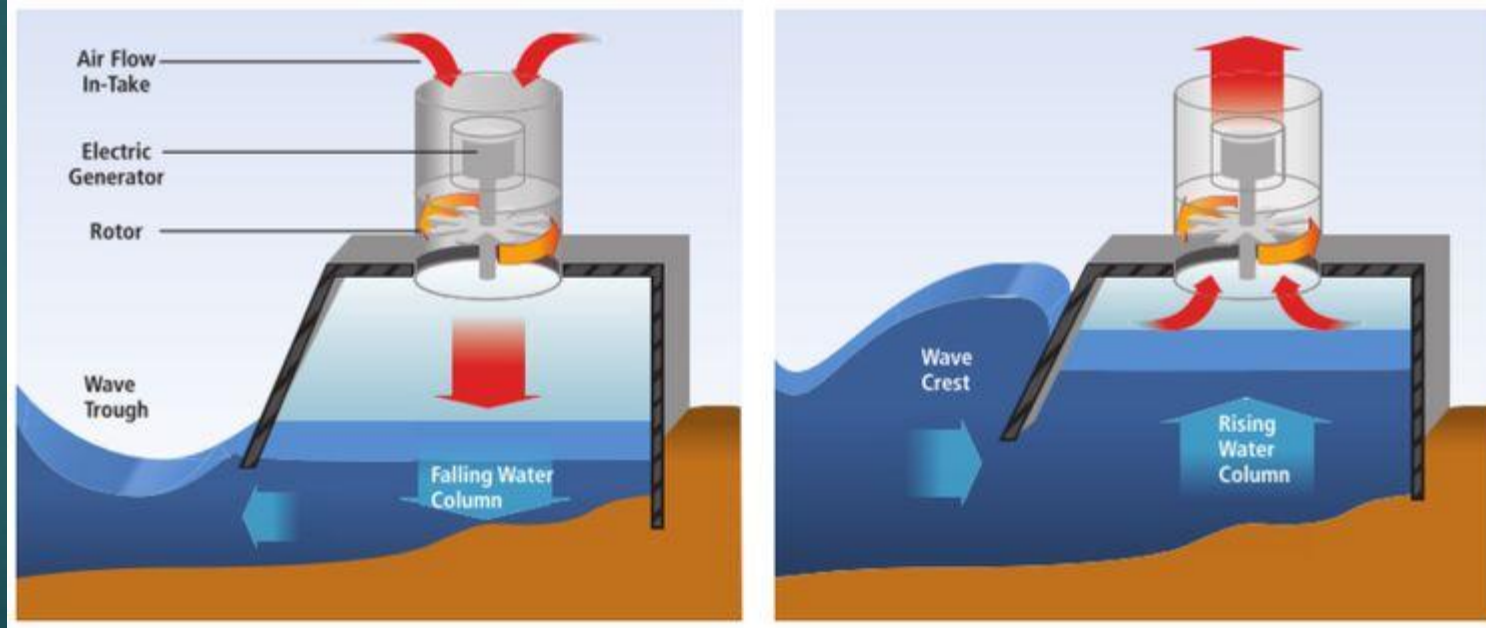
- Definition: Devices that convert the power of ocean waves into usable electricity.
- Global potential: theoretical maximum ~30,000 TWh/yr; exploitable resource $\approx 1/10$ of that.
- Wave energy is more consistent than solar & wind (waves are harvestable ~90% of the time)
- High potential zones: Atlantic coasts (UK, Ireland), Southern Ocean, South Africa, Australia, etc.

Main Types of Wave Energy Converters (WECs)

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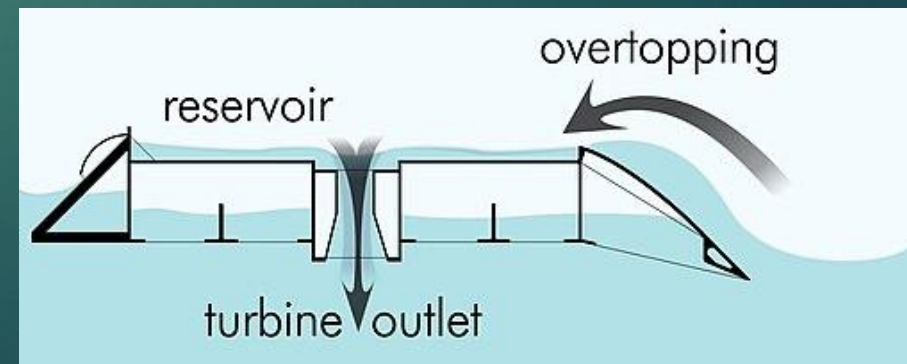
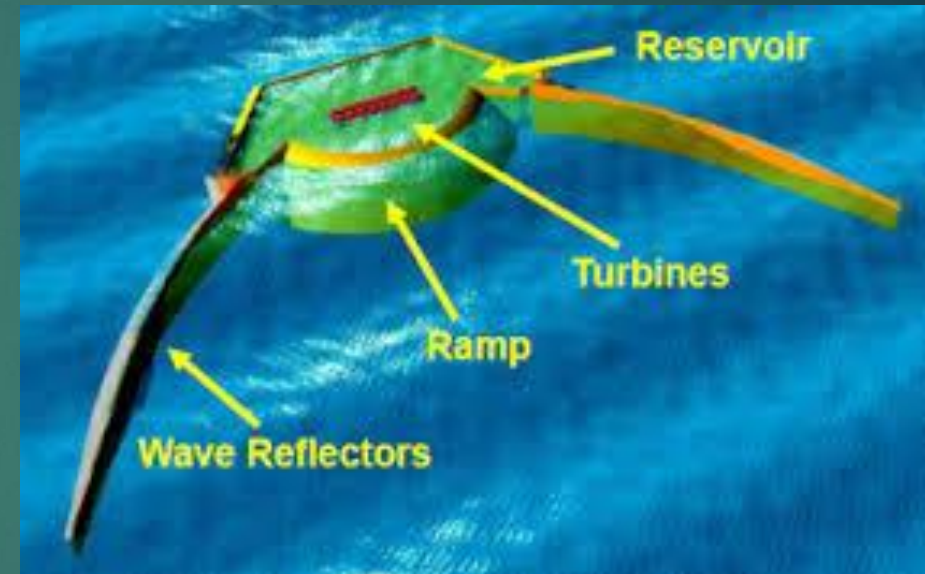
Title: Types of WECs and How They Work

Type	Principle / Example
Oscillating Water Columns (OWC)	Chamber with air above water; wave motion moves water up/down changing pressure; air turbine (e.g., Wells turbine) used.



Overtopping Devices

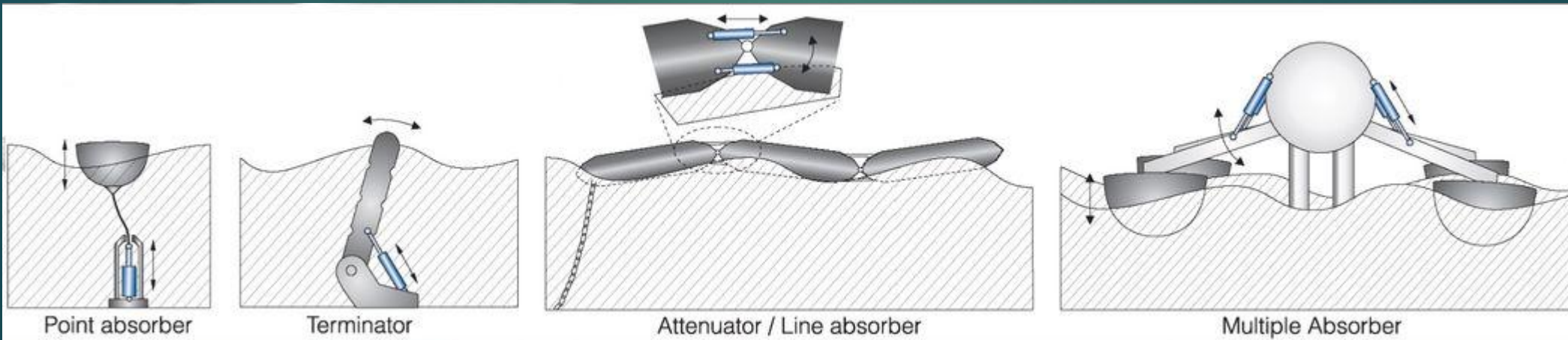
Waves overtop into a reservoir; water flows back to sea via turbines. Example: Wave Dragon.



Wave Absorbing Devices

Includes:

- Heaving Point Absorbers (buoys)
- Oscillating Point Absorbers
- Terminators (flaps)
- Attenuators (long structures riding waves)



Power Take-Off (PTO) Systems & Key Challenges

Title: Converting Motion to Electricity + Challenges

•Power Take-Off Systems (PTO):

- Hydraulic systems, linear or rotary generators, electric drives.
- Some direct drive linear generators; in others air turbines (for OWC), hydro turbines (overtopping).
- Advanced control strategies (phase matching, latching, predictive control) to improve efficiency.

•Challenges:

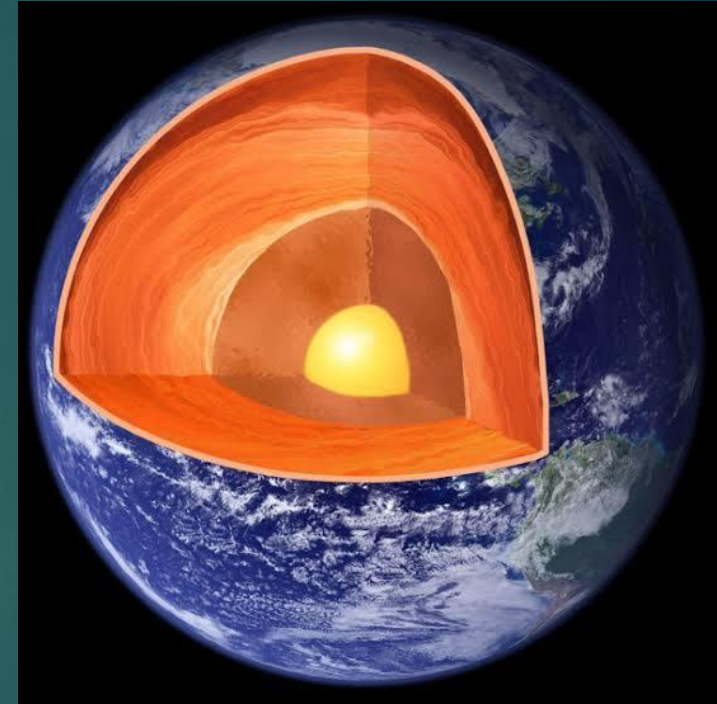
- Harsh marine environment: corrosion, storms, biofouling.
- High capital & maintenance costs; PTO is a large part of total cost.
- Variability and irregularity of waves → makes power output fluctuating.
- Grid integration & mooring issues.

Geothermal energy: Origin, applications, types of geothermal resources.

What is geothermal energy?

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- ▶ Geothermal energy is heat within the earth. The word geothermal comes from the Greek words geo (earth) and therme (heat).
- ▶ Geothermal energy is a renewable energy source because heat is continuously produced inside the earth.
- ▶ People use geothermal heat for bathing, for heating buildings, and for generating electricity.

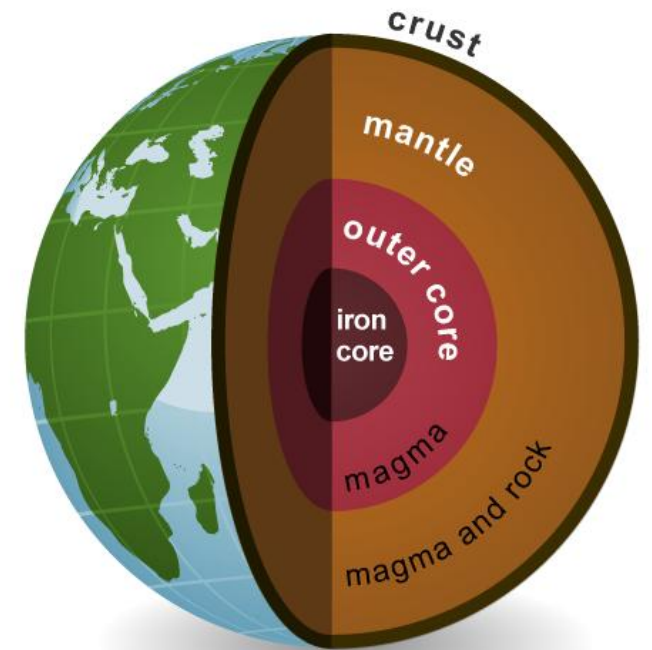


Geothermal Energy & Earth's Heat

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- ▶ **Definition:** Geothermal energy comes from heat inside the Earth, produced by the slow decay of radioactive particles.
- ▶ **Earth's Layers & Temperatures:**
 - ▶ **Inner Core:** Solid iron, $\sim 5,960^{\circ}\text{C}$
 - ▶ **Outer Core:** Molten magma
 - ▶ **Mantle:** Magma & rock, 200°C – $3,990^{\circ}\text{C}$
 - ▶ **Crust:** Solid rock, 5–56 km thick

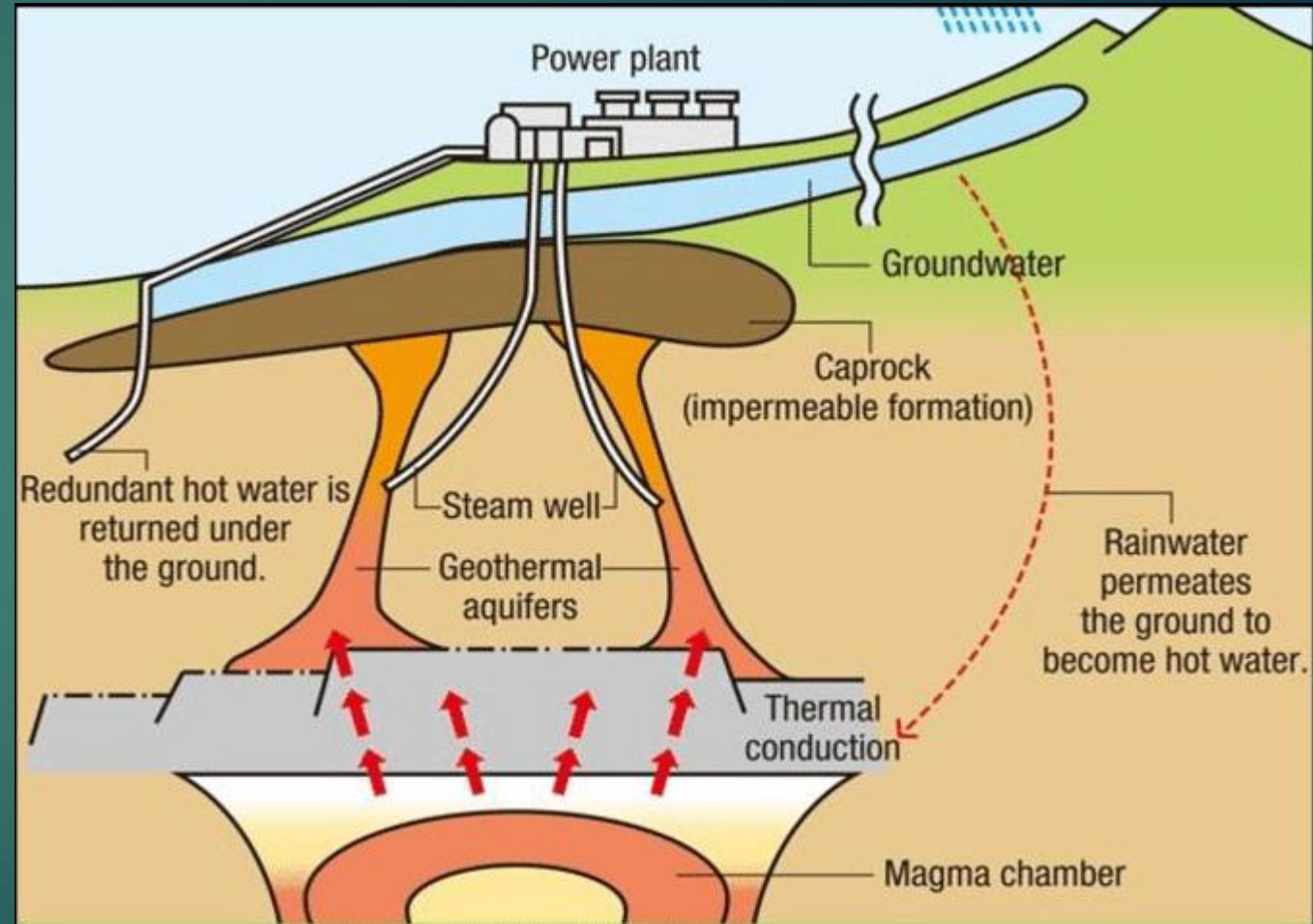
The earth's interior



How Geothermal Energy Reaches the Surface

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- ▶ **Tectonic Plates:** Crust is broken into plates.
- ▶ **Magma Movement:** Magma rises near plate edges through gaps → **volcanoes**.
- ▶ **Lava:** Magma at the surface.
- ▶ **Key Point:** Rocks and water underground absorb heat, which can be harnessed as energy.



Types of Geothermal Resources?

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Primary Classification is Resource Temperature:

- Low Temperature Reservoir: 50-200 °F (10-94 °C)
- High Temperature Reservoir: >200 °F

Low Temperature Reservoirs:

- ▶ Available almost anywhere on earth
- ▶ Predominantly Used for :
 - Heat Pumps , Space Heating
- ▶ Other Common Uses:
 - Hot Water Production
 - Piped Under Roads / Sidewalks (Klamath Falls, Oregon) – Snow melting.
 - In Greenhouses to Grow Flowers - using hot well water.
 - Industrial Uses: dry wood, pasteurize milk, grow fish, etc.

Applications

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High Temperature Reservoirs:

- ▶ Availability:
 - Can Occur Within a Couple of Miles of Earth's Surface
Where Earth's Crust Is Very Thin – i.e., Closer to Molten Magma at Core
- ▶ Suitable for Commercial Production of Electricity
 - Power Plants Need High Capacity Geothermal Reservoir
– Water / Steam $>220^{\circ}\text{F}$ (105°C)
- ▶ Greatest Potential for Energy Output

Basic Types of Geothermal Reservoirs

3 General Classes of Geothermal Uses

- ▶ (1) Ground Source Heat Pump (Low Temperature)
- ▶ (2) Electricity Generation: Power Plants (Steam >200F)

(1) Ground Source Heat Pump

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- ▶ Geothermal heat pumps (GHPs), also called ground-source heat pumps, are energy-efficient systems that use the stable temperature of the earth to provide heating and cooling for buildings, serving as a heat source in winter and a heat sink in summer.

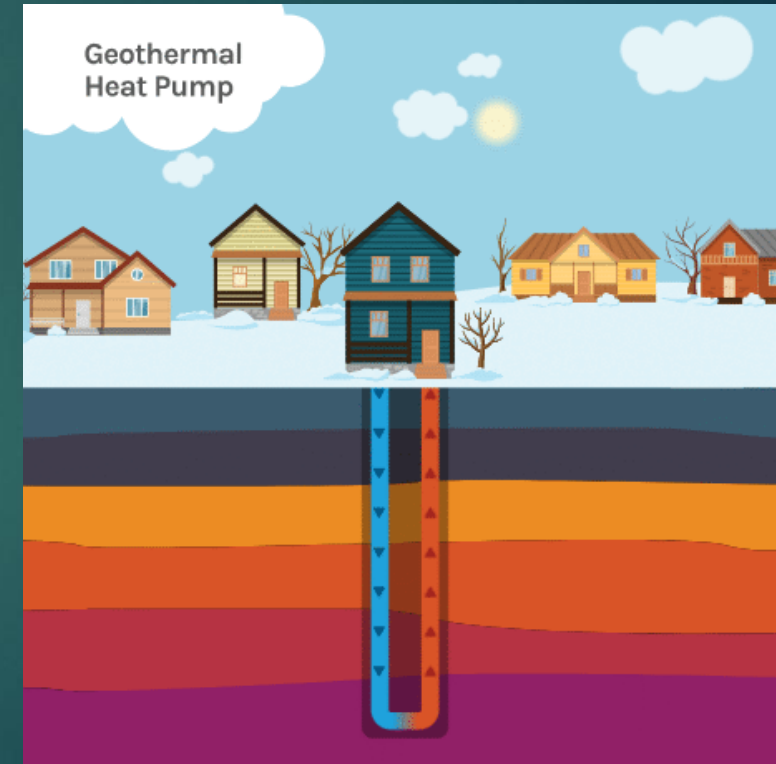
How They Work:

Heat Transfer: Geothermal heat pumps leverage the constant temperature of the shallow earth, which is typically warmer than the air in winter and cooler in summer.

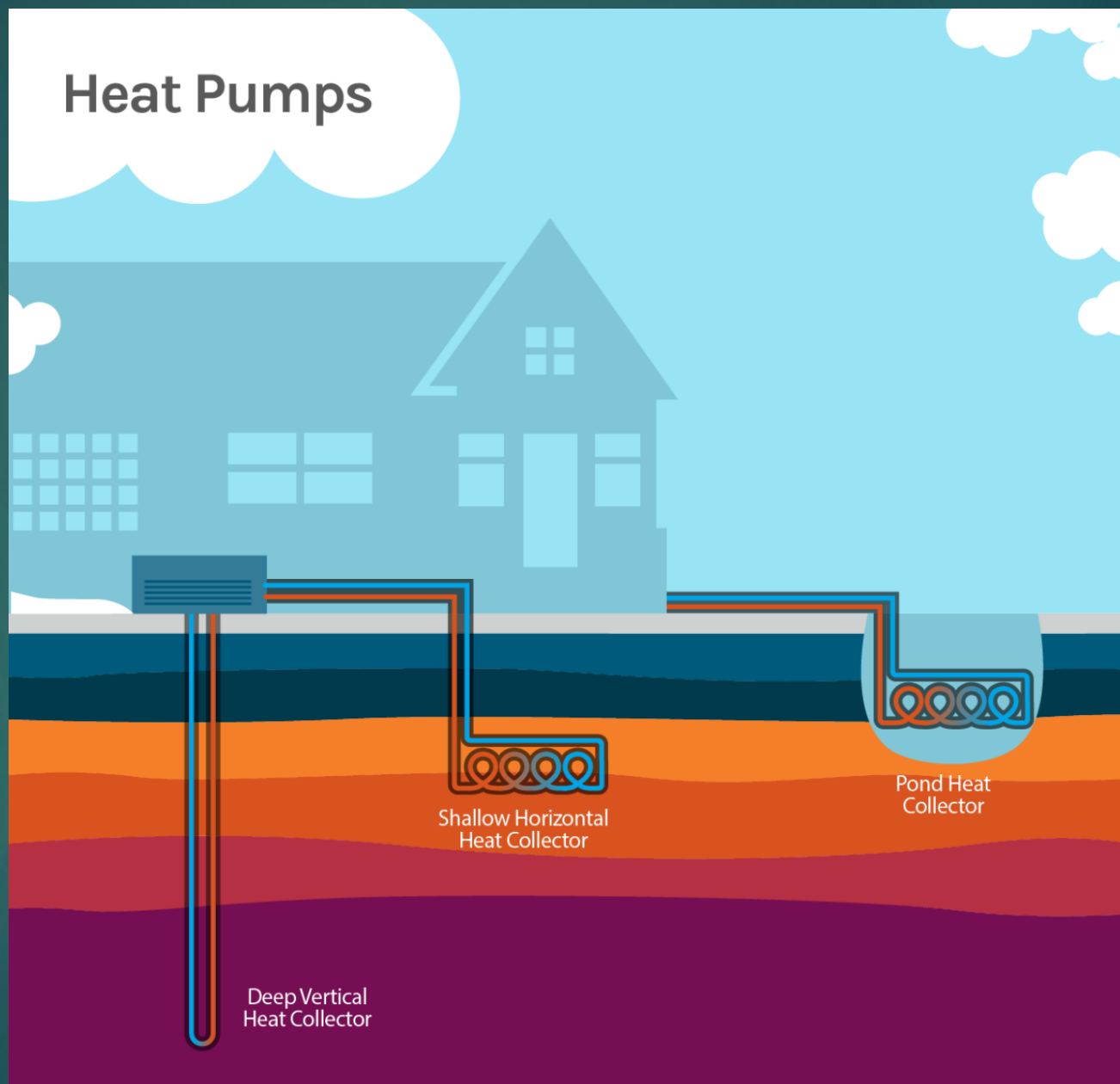
Summer (Cooling): GHPs extract heat from a building and transfer it to the ground, using the earth as a heat sink.

Winter (Heating): The system absorbs heat from the ground and transfers it into the building to provide warmth.

Earth Loops: These systems involve a network of buried plastic pipes (an "earth loop") through which a fluid circulates, transferring heat to or from the surrounding soil or water.



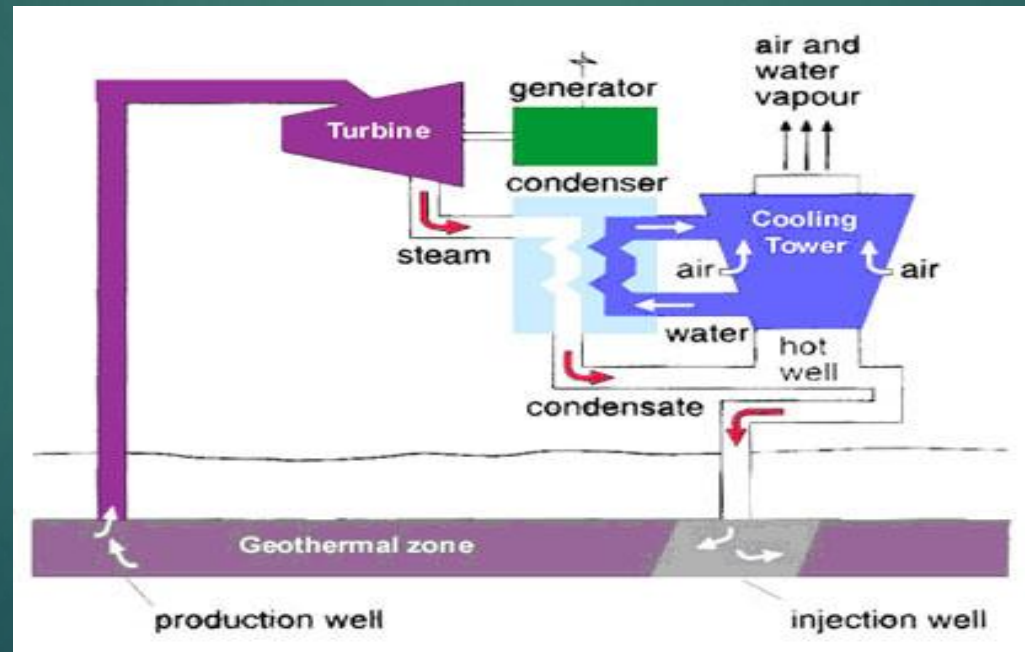
Heat Pumps

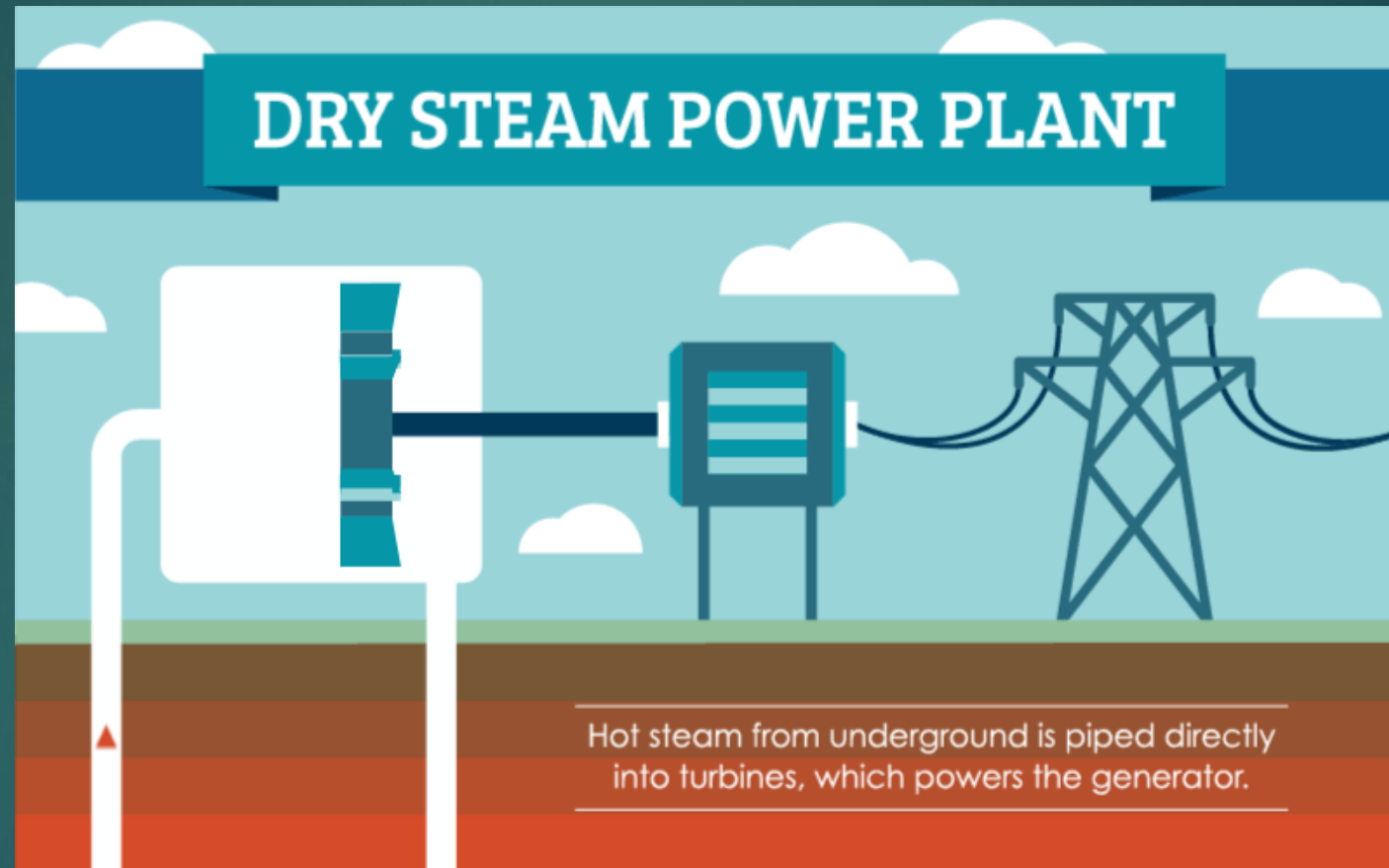


(2) Electricity Generation

(a) Dry Steam Power Plant:

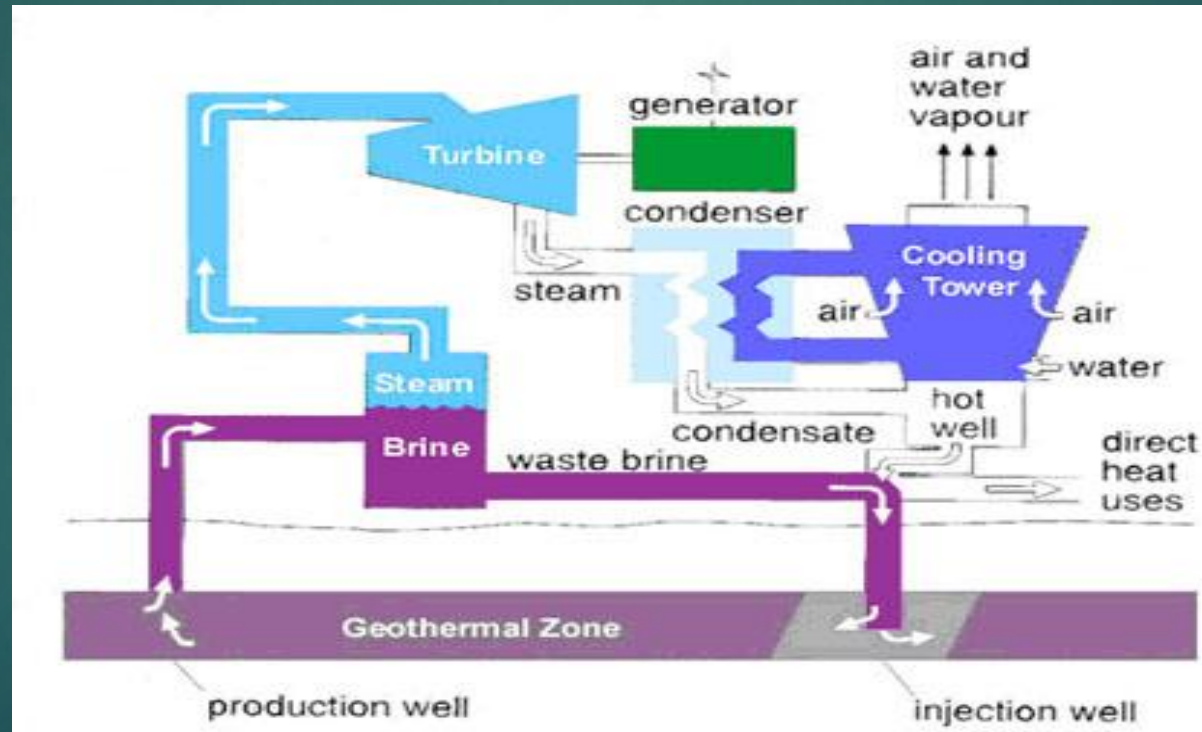
- ▶ Dry steam plants use steam directly from a geothermal reservoir to turn generator turbines.
- ▶ Uses the superheated, pressurized steam (180° - 350°C)

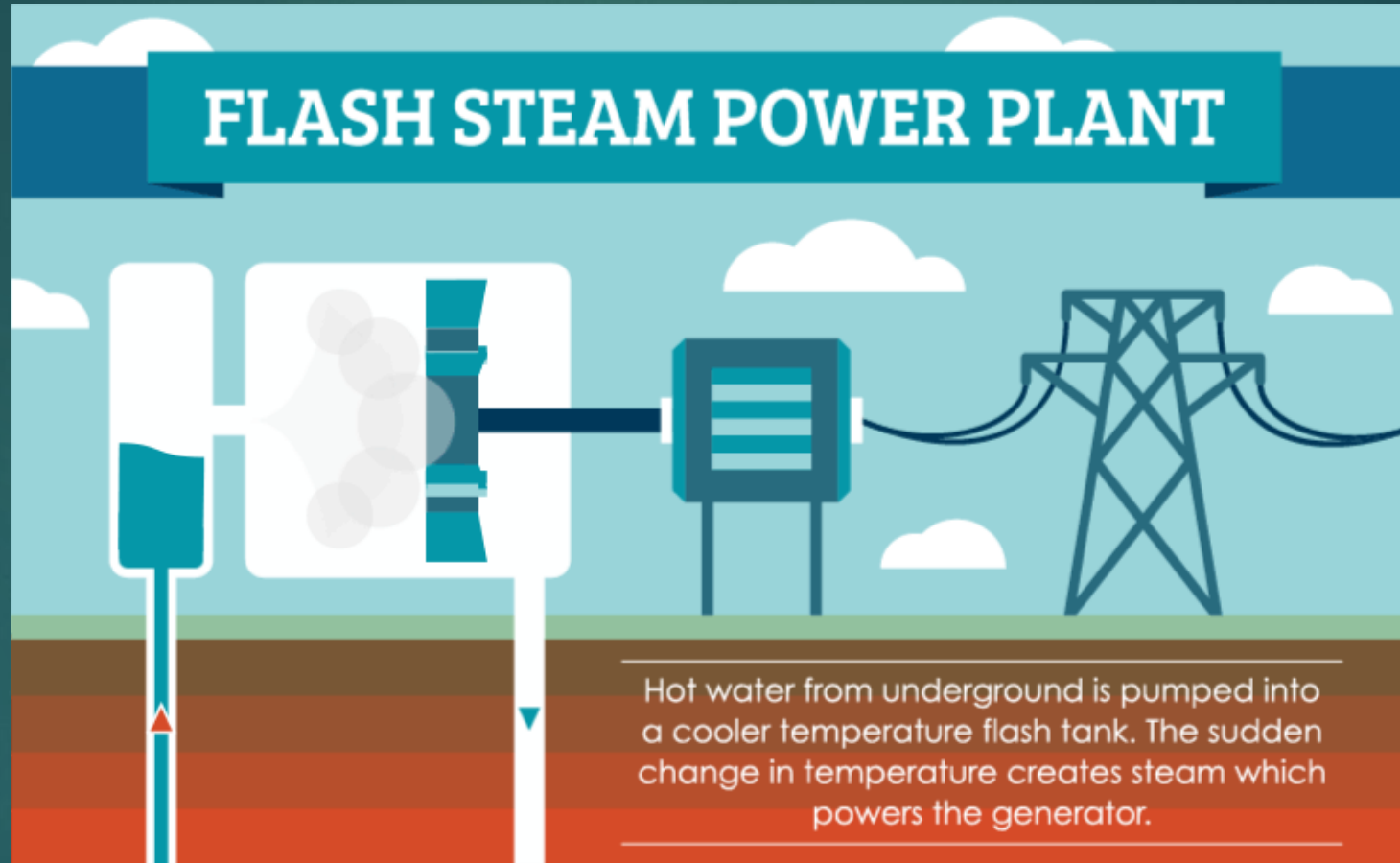




(b) Flash Steam Power Plant:

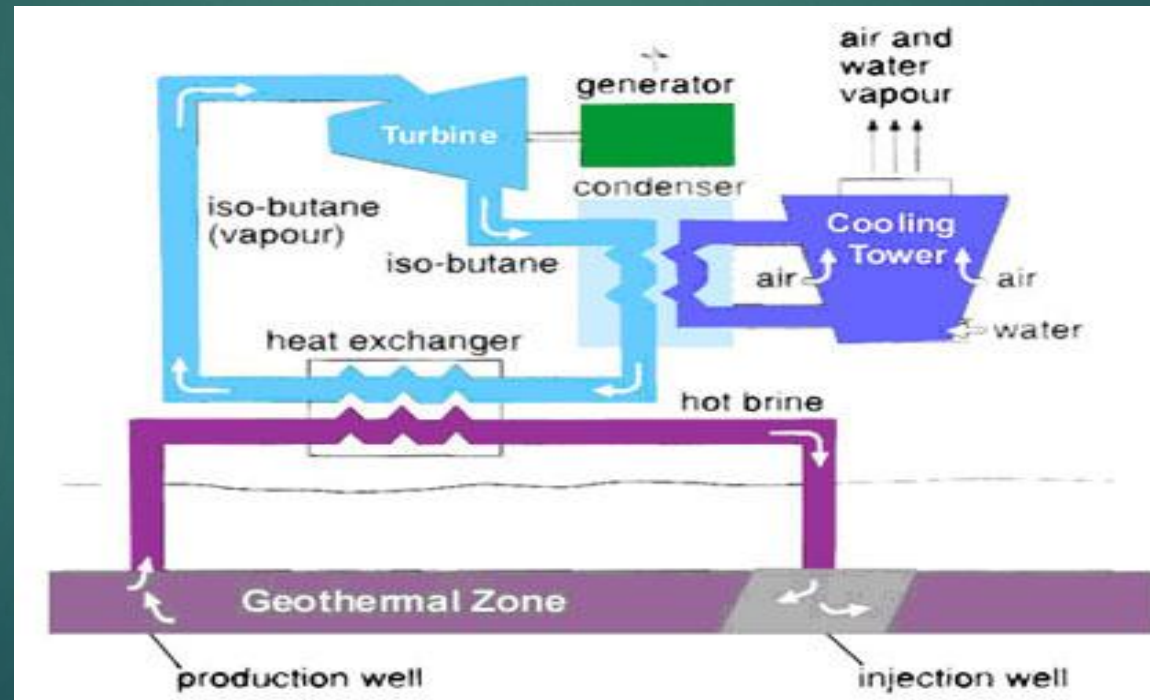
- ▶ Uses high-temperature (over 182°C / 360°F) hot water from deep underground
- ▶ Water is brought to the surface under high pressure, where the pressure is then reduced, causing some of the water to "flash" into steam.

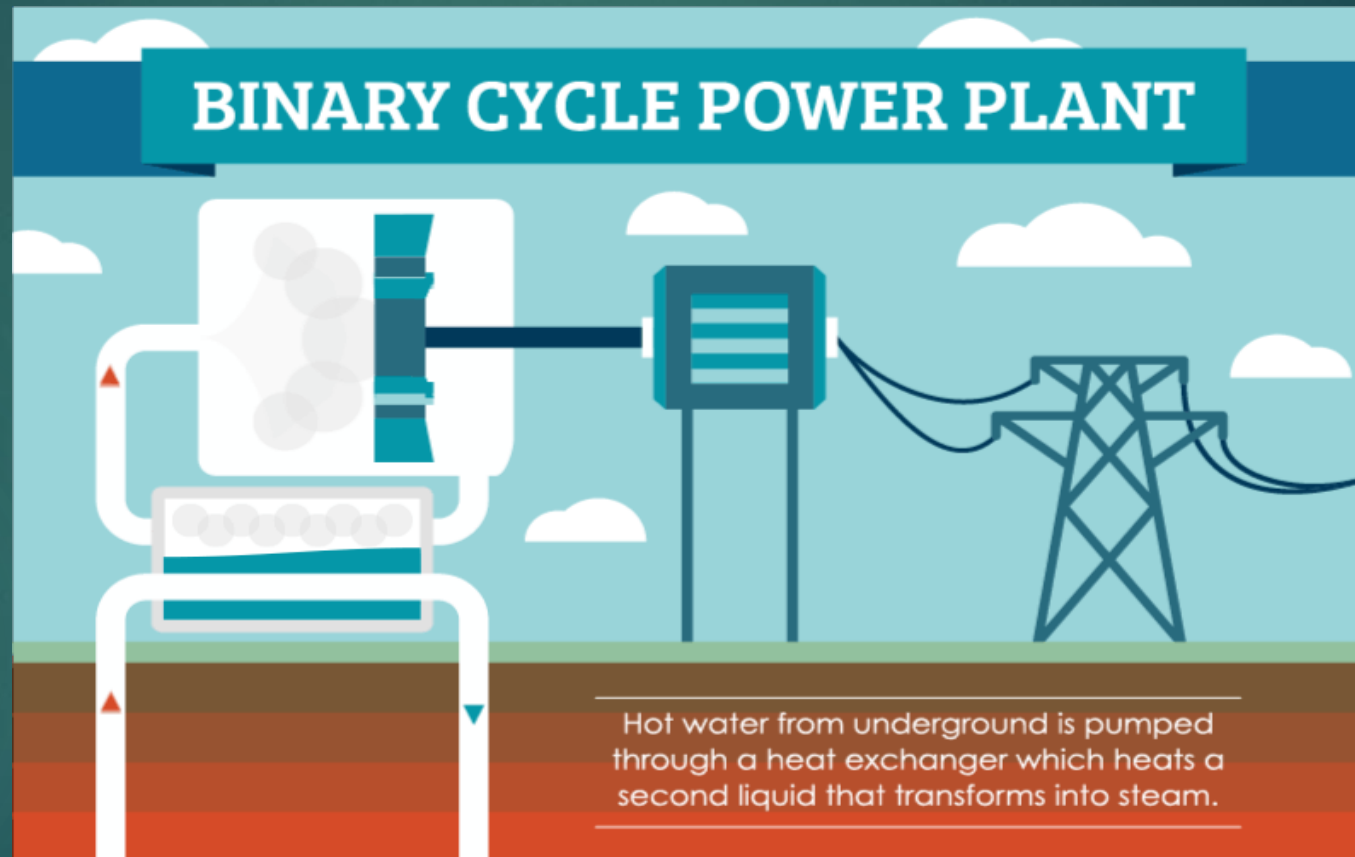




(C) Binary Cycle Power Plant:

- ▶ Uses the heat from a lower-temperature resource to **vaporize a secondary organic working fluid** with a low boiling point in a heat exchanger
- ▶ This vapor drives a turbine that generates electricity, after which the working fluid is condensed and pumped back to the heat exchanger to repeat the cycle.





Geothermal Energy in INDIA

<i>S.N</i>	<i>Geothermal provinces</i>	<i>Main Locations</i>	<i>Heat flow (mW/m²)</i>	<i>Thermal gradient (°C/km)</i>
1.	Himalayas	Puga Valley (J and K), Manikaran (HP)	468	234
2.	Sohana			76–96
3.	West coast	Unai (Maharashtra)	129	59
4.	Cambay	Tuwa (Gujrat)	93	70
5.	Son–Narmada– Tapi (SONATA)	Jalgaon (Maharashtra), Tattapani (Chattisgarh), Bakreswar (Bihar)	120–260	60–90
6.	Godavari		104	60
7.	Mahanadi		200	90

Unit-4 - Biomass

- ▶ *Biomass, Sources of biomass, Pyrolysis, combustion and gasification process, Updraft and downdraft gasifier, Fluidized bed gasifier, Fermentation and digestion process, Fixed and floating digester biogas plants, Operation of biogas plants, Applications.*